

TACHS Full-Length Practice Exam 07

Reading · Written Expression · Mathematics · Ability

This printable contains 200 multiple-choice questions arranged in four parts, followed by the validated answer keys and explanations supplied with each part.

Recommended section times: Reading 40 minutes · Written Expression 40 minutes · Mathematics 45 minutes · Ability 35 minutes

Exam Part A

Filename: tachs_full_exam_07_part_a.txt

Exam Family: TACHS

Exam Number: 7

Exam Part: A

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R7-S01

Section: Reading

Stimulus Type: Literary Passage

Title: The Grid Beneath the Rain

Text:

Leah had spent Saturday drawing chalk guidelines for a mural on the wall beside the community basketball court. Each square in her paper sketch corresponded to a larger chalk square on the wall. On Sunday morning, a brief shower washed away nearly every line.

The painting team would arrive in an hour. Leah considered redrawing the entire grid, but the wall was still damp and chalk would not hold. As she looked closely, she noticed that the concrete blocks formed rows of equal rectangles. Their seams were not the same size as her planned squares, yet they created a permanent pattern.

Leah placed small pieces of removable tape at selected seams. She revised her sketch so that every two blocks represented one grid unit. A few curved parts of the design shifted slightly, but the main proportions remained. When the other painters arrived, they used the block seams to locate each section.

By afternoon, the mural covered both the tape and the evidence of Leah's first grid. Only the straight joints between blocks remained visible. Instead of distracting from the painting, the joints made the background appear textured. Leah had thought the wall was simply a surface waiting for her plan. The rain forced her to notice that the wall already had a structure of its own.

STIMULUS END

TACHS-R7-001

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S01

Prompt:

What destroys Leah's original chalk grid?

- A) A brief rain shower
- B) The basketball team
- C) Wet paint from the mural
- D) Removable tape

TACHS-R7-002

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S01

Prompt:

Why does Leah revise her paper sketch?

- A) She decides to paint a completely different subject.
- B) She needs the design to correspond with the wall's block seams.
- C) She wants every painter to use a separate scale.
- D) She discovers that the wall is too small for any mural.

TACHS-R7-003

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S01

Prompt:

Which statement best expresses a central idea of the passage?

- A) Recognizing an existing structure can lead to an effective adaptation.
- B) Outdoor murals are impossible to complete after rain.
- C) Careful plans should never be changed after work begins.
- D) A textured wall always produces a better painting than a smooth wall.

TACHS-R7-004

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S01

Prompt:

As used in the third paragraph, "proportions" most nearly means

- A) paint colors.
- B) amounts of tape.
- C) relative sizes of the design's parts.
- D) working hours.

STIMULUS START

Stimulus ID: TACHS-R7-S02

Section: Reading

Stimulus Type: Informational Passage

Title: Helping Fish Pass a Dam

Text:

A dam can slow water for power generation, irrigation, or flood control, but it may also block fish that travel upstream to reach spawning habitat. Fish passages are structures designed to provide another route around or through the barrier.

One common design, often called a fish ladder, consists of a series of ascending pools. Water flows from one pool to the next through openings or over low walls. Instead of making one impossible leap over the full height of a dam, a fish swims through shorter sections and rests in calmer pools between them.

The design must match local species and conditions. Some fish can swim through fast water or jump small barriers, while others need gentler currents. Engineers consider water depth, flow speed, seasonal river levels, and the locations fish naturally search for a route. If the entrance is difficult to find, even a well-built passage may be ineffective.

Researchers monitor passages with counters, cameras, or tagged fish. These observations can reveal whether fish enter, where they pause, and whether certain species turn back. A passage is therefore not merely installed and forgotten. Its performance must be measured, and its flow or entrance may need adjustment as conditions change.

STIMULUS END

TACHS-R7-005

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S02

Prompt:

What feature of a fish ladder provides resting places?

- A) A dry channel beside the dam
- B) A single tall wall
- C) A tagged camera
- D) A series of calmer pools

TACHS-R7-006

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S02

Prompt:

Why does the author mention that fish differ in swimming and jumping ability?

- A) To argue that all dams should have identical ladders
- B) To explain why one passage design may not suit every species
- C) To show that river levels never affect fish movement
- D) To prove that cameras change how fish swim

TACHS-R7-007

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R7-S02

Prompt:

How is the passage mainly organized?

- A) A problem is introduced, a design is explained, and factors affecting its success are discussed.
- B) Several dams are ranked from oldest to newest.
- C) A scientist's biography is followed by a personal opinion.
- D) Two fish species debate which route to use.

TACHS-R7-008

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S02

Prompt:

Which observation would most strongly suggest that a passage entrance needs adjustment?

- A) Many fish enter and continue upstream.
- B) Water depth remains within the planned range.
- C) Cameras record fish resting in several pools.
- D) Tagged fish repeatedly approach the dam but fail to enter the passage.

STIMULUS START

Stimulus ID: TACHS-R7-S03

Section: Reading

Stimulus Type: Humanities Passage

Title: The Work of a Replica

Text:

A museum replica may look like a substitute for an original object, but a well-made copy can serve purposes the original cannot. A fragile sculpture may need to remain behind glass, while a replica can be handled to study its weight, texture, or construction.

Copies also make comparisons possible. Students can place reproductions of two carved tools side by side, rotate them, and test how each fits the hand. Such activity may reveal a design difference that is difficult to notice in photographs. For visitors with limited vision, a touchable replica can provide access to shape and surface that a label alone cannot offer.

Replicas must be identified honestly. A copy should not be presented as the historical object itself, and makers should explain which details are based on evidence and which have been reconstructed. The materials used in a modern replica may also behave differently from the original materials.

These limitations do not make copies useless. Instead, they define the questions a replica can answer. An original carries direct evidence of its history; a replica allows forms of examination that might damage that evidence. When displayed together and clearly labeled, original and copy can offer different, complementary kinds of understanding.

STIMULUS END

TACHS-R7-009

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S03

Prompt:

According to the passage, how can a replica assist a visitor with limited vision?

- A) It can replace every written label.
- B) It can prove the exact age of the original.
- C) It can be touched to explore shape and surface.
- D) It can guarantee that reconstructed details are correct.

TACHS-R7-010

Section: Reading

Type: mcq
Category: Reading
Skill: Vocabulary in Context
Stimulus ID: TACHS-R7-S03
Prompt:

As used in the final paragraph, “complementary” most nearly means

- A) identical in every detail.
- B) made from the same historical material.
- C) too fragile to be examined.
- D) useful in different ways that work together.

TACHS-R7-011

Section: Reading
Type: mcq
Category: Reading
Skill: Main Idea, Central Thought, or Summary
Stimulus ID: TACHS-R7-S03
Prompt:

Which statement best summarizes the passage?

- A) Clearly identified replicas can support hands-on study while originals preserve direct historical evidence.
- B) Modern copies contain more historical evidence than originals.
- C) Replicas should replace fragile originals in every display.
- D) Photographs provide the only reliable way to compare objects.

TACHS-R7-012

Section: Reading
Type: mcq
Category: Reading
Skill: Interpretative Comprehension or Inference
Stimulus ID: TACHS-R7-S03
Prompt:

Why should makers explain which replica details were reconstructed?

- A) So visitors can distinguish evidence-based features from informed additions
- B) So the replica can be sold as an original object
- C) So visitors will avoid comparing it with photographs
- D) So every copy will be made from ancient material

TACHS-R7-013

Section: Reading
Type: mcq
Category: Reading
Skill: Evaluative Comprehension, Author's Purpose, or Tone
Stimulus ID: TACHS-R7-S03
Prompt:

The author’s attitude toward museum replicas is best described as

- A) dismissive because copies lack all value.
- B) enthusiastic without recognizing limitations.

C) supportive when replicas are used transparently and for suitable purposes.

D) uncertain about whether museums should use them.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE7-S01

Section: Written Expression

Stimulus Type: Revision Passage

Title: A Cart for Shared Books

Text:

[1] Our recreation center has books in several rooms, but visitors often do not know where to return them. [2] We propose placing a rolling book cart near the main desk. [3] One shelf would hold books ready to borrow, while another would collect returns. [4] Every Friday, volunteers will sort returned books and move them to the correct rooms. [5] The cart will have large labels, it will also include a clipboard for suggesting new titles. [6] Its wheels should lock so the cart remains steady while children browse. [7] The center's front doors are made of glass. [8] A six-week trial will show whether the cart makes borrowing and returning books easier.

STIMULUS END

TACHS-WE7-001

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S01

Prompt:

Which sentence best introduces the proposed solution?

A) Sentence 1

B) Sentence 4

C) Sentence 2

D) Sentence 8

TACHS-WE7-002

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S01

Prompt:

Which revision corrects the error in sentence 5?

A) Having large labels, and a clipboard for suggesting new titles.

B) The cart will have large labels it will also include, a clipboard for suggesting new titles.

C) The cart will have large labels; it will also include a clipboard for suggesting new titles.

D) The cart will have large labels, because also a clipboard for titles.

TACHS-WE7-003

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S01

Prompt:

Which revision of sentence 4 is most concise?

- A) On every Friday of each week, volunteers will do the sorting of books that have been returned.
- B) Each Friday, volunteers will sort returns and deliver them to the appropriate rooms.
- C) Volunteers on Friday will sort and move all of the books to where the books should go in rooms.
- D) Returned books will be sorted by volunteers and moved in a correct sort of way.

TACHS-WE7-004

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S01

Prompt:

Which sentence should be removed because it is irrelevant?

- A) Sentence 3
- B) Sentence 6
- C) Sentence 8
- D) Sentence 7

STIMULUS START

Stimulus ID: TACHS-WE7-S02

Section: Written Expression

Stimulus Type: Revision Passage

Title: Making Windows Safer for Birds

Text:

[1] Our environmental club studied bird collisions with the windows near the school courtyard. [2] For two weeks, we checked the ground each morning and recorded where evidence of a collision appeared. [3] Most incidents occurred at three windows that reflected trees. [4] We applied small dot markers to the outside surface of those windows in a closely spaced pattern. [5] The dots makes the glass more visible to approaching birds while preserving light indoors. [6] During the following two weeks, we continued the same morning checks. [7] We recorded fewer incidents, but a longer observation period is needed before reaching a firm conclusion. [8] The project taught us that a simple change should still be evaluated with consistent evidence.

STIMULUS END

TACHS-WE7-005

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE7-S02

Prompt:

Which revision corrects the error in sentence 5?

- A) The dots makes the glass more visible to approaching birds while preserving light indoors.
- B) The dots make the glass more visible to approaching birds while preserving light indoors.
- C) The dots has made the glass more visible to approaching birds while preserving light indoors.
- D) The dots making the glass more visible to approaching birds while preserving light indoors.

TACHS-WE7-006

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S02

Prompt:

Which revision most effectively combines sentences 6 and 7?

- A) During the following two weeks, checking mornings, fewer incidents but needing a conclusion later.
- B) During the following two weeks, we continued the same morning checks and recorded fewer incidents, although more observation is needed before a firm conclusion.
- C) We checked, the incidents were fewer, a longer period before concluding.
- D) Although a longer observation period, during two weeks and the checks continued.

TACHS-WE7-007

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S02

Prompt:

Which phrase best replaces “where evidence of a collision appeared” in sentence 2?

- A) the area of evidence and things
- B) where some stuff had happened
- C) places that looked like something occurred
- D) the location of each suspected collision

TACHS-WE7-008

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S02

Prompt:

Which sentence would best follow sentence 3?

- A) The courtyard also contains four wooden benches.
- B) Our club members prefer meeting after school.
- C) Birds live on every continent except Antarctica.
- D) These windows became the focus of our first test.

TACHS-WE7-009

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The news from both teams were encouraging.
- B) Mathematics are my first class of the day.
- C) The pair of gloves is in the top drawer.
- D) The basket of apples have fallen.

TACHS-WE7-010

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: None

Prompt:

Which sentence uses parallel structure correctly?

- A) The internship required arriving early, taking careful notes, and asking questions.
- B) The internship required early arrival, taking careful notes, and questions.
- C) The internship required us to arrive early, careful notes, and asking questions.
- D) The internship required arriving early, careful notes, and to ask questions.

TACHS-WE7-011

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: None

Prompt:

Which sentence expresses the distribution most clearly and correctly?

- A) The two sisters shared the prize among a group of two.
- B) Please divide the four notebooks among the students.
- C) The teacher placed the papers between the entire class.
- D) Among the two final choices, I prefer the first.

TACHS-WE7-012

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The storm ended shortly before dawn.

Line 2: Several branches lay across the road.

Line 3: Workers moved the branches too the shoulder.

Line 4: Traffic resumed before the morning rush.

- A) Line 3
- B) Line 2
- C) Line 1

D) Line 4

TACHS-WE7-013

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: A flock of geese crossed the field.

Line 2: The birds paused beside a shallow pond.

Line 3: One goose lifted its head and called loudly.

Line 4: Soon, the entire flock moved more farther north.

A) Line 1

B) Line 4

C) Line 3

D) Line 2

SECTION 3 - MATHEMATICS

TACHS-M7-001

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is $\frac{7}{12} + \frac{5}{18}$?

A) $\frac{11}{30}$

B) $\frac{31}{36}$

C) $\frac{12}{30}$

D) $1\frac{1}{36}$

TACHS-M7-002

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $6x - 11 = 31$.

A) 7

B) 5

C) 8

D) 10

TACHS-M7-003

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The values are 9, 12, 15, 15, 19, and 26. What is the mode?

- A) 12
- B) 14
- C) 15
- D) 16

TACHS-M7-004

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

An isosceles triangle has a vertex angle of 44° . What is the measure of each equal base angle?

- A) 44°
- B) 58°
- C) 136°
- D) 68°

TACHS-M7-005

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A bottle contains 1.8 liters. How many 300-milliliter servings does it contain?

- A) 4
- B) 5
- C) 6
- D) 9

TACHS-M7-006

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which number is divisible by 9?

- A) 4,318

- B) 7,125
- C) 6,431
- D) 5,472

TACHS-M7-007

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

A pattern follows 3, 8, 18, 38. Which rule produces each term after the first?

- A) Add 5.
- B) Multiply by 2, then add 2.
- C) Multiply by 2.
- D) Add consecutive multiples of 5.

TACHS-M7-008

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 2 yellow, 6 purple, and 4 orange tiles. What is the probability of selecting yellow or orange?

- A) $\frac{1}{3}$
- B) $\frac{5}{6}$
- C) $\frac{2}{3}$
- D) $\frac{1}{2}$

TACHS-M7-009

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangle has area 96 square units and width 8 units. What is its perimeter?

- A) 40 units
- B) 20 units
- C) 32 units
- D) 48 units

TACHS-M7-010

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A train travels 156 miles in 3 hours at a constant rate. How far does it travel in 5 hours?

- A) 208 miles
- B) 260 miles
- C) 312 miles
- D) 240 miles

TACHS-M7-011

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

A \$150 item is increased in price by 12%. What is the new price?

- A) \$168
- B) \$172
- C) \$162
- D) \$180

TACHS-M7-012

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Which expression represents five more than the quotient of p and 4?

- A) $p/4 + 5$
- B) $5p/4$
- C) $p/(4 + 5)$
- D) $(p + 5)/4$

SECTION 4 - ABILITY

TACHS-A7-001

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the filled shape moves from the left of the open shape to its right.

Row 1: ● ○ | ○ ●

Row 2: ■ □ | ?

A) ☐ ☐

B) ☒ ☐

C) ☒ ☒

D) ☐ ☒

TACHS-A7-002

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, one vertical line is added.

Top-left: |

Bottom-left: ||

Top-right: ||

Bottom-right: ?

A) No line

B) |

C) ||||

D) |||

TACHS-A7-003

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the third cell contains the first symbol followed by the second symbol twice.

Row 1: ○ | □ | ○ □ □

Row 2: △ | + | ?

A) △ + +

B) + + △

C) + △ +

D) △ △ +

TACHS-A7-004

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the arrow reverses direction and the symbol after it changes from + to ×.

Row 1: ↑ + | ↓ ×

Row 2: → + | ?

A) ← ×

B) → ×

C) ← +

D) ↑ ×

TACHS-A7-005

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell keeps only the symbols appearing in both left groups.

Row 1: {○, □, △} and {□, △, +} | {□, △}

Row 2: {★, +, ×} and {+, ×, ◇} | ?

A) {★, ◇}

B) {+, ×}

C) {★, +}

D) {×, ◇}

TACHS-A7-006

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square paper is folded from right to left. A hole is punched near the folded paper's bottom edge, halfway between its left and right sides and away from the fold. What appears after unfolding?

A) Two holes in a horizontal pair near the bottom

B) Four holes near the bottom corners

C) Two holes in a vertical pair

D) One hole on the bottom edge

TACHS-A7-007

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded from bottom to top. Two holes are punched at different horizontal positions, both away from edges. What appears after unfolding?

A) Three holes around the center

B) Four holes forming two vertical pairs

- C) Two holes in one row
- D) Eight holes in four pairs

TACHS-A7-008

Section: Ability
 Type: mcq
 Category: Ability
 Skill: Paper Folding
 Stimulus ID: None
 Prompt:

A square is folded left to right and then top to bottom. A circular cut is made at the corner of the folded square formed by the two original outer edges. What appears after unfolding?

- A) One circular opening at the center
- B) Two openings on one edge
- C) Four openings meeting at the center
- D) Four quarter-circle cuts, one at each outer corner

TACHS-A7-009

Section: Ability
 Type: mcq
 Category: Ability
 Skill: Paper Folding
 Stimulus ID: None
 Prompt:

A square is folded once along the diagonal from upper-right to lower-left. A hole is punched exactly on the folded diagonal, away from corners. What appears after unfolding?

- A) Eight holes along the diagonal
- B) Two holes mirrored across the diagonal
- C) One hole on the diagonal
- D) Four holes in a square

TACHS-A7-010

Section: Ability
 Type: mcq
 Category: Ability
 Skill: Figure Classification
 Stimulus ID: None
 Prompt:

The following groups share one rule:

1. ○ ● ●
2. □ ■ ■
3. △ ▲ ▲

Which choice follows the same rule?

- A) ◆ ◇ ◇
- B) ○ ○ ●
- C) ☆ ★ ☆
- D) ◇ ◆ ◆

TACHS-A7-011

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. Top ○; middle +; bottom □
2. Top △; middle +; bottom ◇
3. Top ☆; middle +; bottom ○

Which choice follows the same rule?

- A) Top ■; middle +; bottom □
- B) Top ◇; middle ×; bottom △
- C) Top ○; middle +; bottom ●
- D) Top □; middle +; bottom △

TACHS-A7-012

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following pairs share one rule:

1. | becomes -
2. / becomes \
3. - becomes |

Which choice follows the same rule?

- A) + becomes ×
- B) | becomes /
- C) \ becomes /
- D) / becomes |

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R7-001 - Correct: A - Correct Answer: A brief rain shower - Explanation: The passage states that a brief shower washes away nearly every chalk line.

TACHS-R7-002 - Correct: B - Correct Answer: She needs the design to correspond with the wall's block seams. - Explanation: Revising the sketch lets Leah use two concrete blocks as each new grid unit.

TACHS-R7-003 - Correct: A - Correct Answer: Recognizing an existing structure can lead to an effective adaptation. - Explanation: Leah succeeds by adapting her plan to the permanent structure already present in the wall.

TACHS-R7-004 - Correct: C - Correct Answer: relative sizes of the design's parts. - Explanation: The revised grid preserves the relative sizes and relationships among the mural's main parts.

TACHS-R7-005 - Correct: D - Correct Answer: A series of calmer pools - Explanation: The ascending pools include calmer areas where fish can rest between shorter swimming sections.

TACHS-R7-006 - Correct: B - Correct Answer: To explain why one passage design may not suit every species - Explanation: Species differences explain why passage speed, depth, and barriers must be designed for local fish.

TACHS-R7-007 - Correct: A - Correct Answer: A problem is introduced, a design is explained, and factors affecting its success are discussed. - Explanation: The passage moves from the barrier problem to passage design, customization, and monitoring.

TACHS-R7-008 - Correct: D - Correct Answer: Tagged fish repeatedly approach the dam but fail to enter the passage. - Explanation: Repeated approaches without entry indicate that fish may not locate or access the entrance effectively.

TACHS-R7-009 - Correct: C - Correct Answer: It can be touched to explore shape and surface. - Explanation: The passage specifically notes that touchable replicas provide access to shape and surface.

TACHS-R7-010 - Correct: D - Correct Answer: useful in different ways that work together. - Explanation: The original and replica contribute distinct but mutually useful forms of understanding.

TACHS-R7-011 - Correct: A - Correct Answer: Clearly identified replicas can support hands-on study while originals preserve direct historical evidence. - Explanation: The passage explains the different, complementary roles of originals and honestly labeled replicas.

TACHS-R7-012 - Correct: A - Correct Answer: So visitors can distinguish evidence-based features from informed additions - Explanation: Explaining reconstructed details helps viewers understand the limits of what the copy can demonstrate.

TACHS-R7-013 - Correct: C - Correct Answer: supportive when replicas are used transparently and for suitable purposes. - Explanation: The author values replicas while emphasizing honest labeling, material limits, and appropriate questions.

WRITTEN EXPRESSION ANSWERS

TACHS-WE7-001 - Correct: C - Correct Answer: Sentence 2 - Explanation: Sentence 2 states the proposal directly after sentence 1 explains the return problem.

TACHS-WE7-002 - Correct: C - Correct Answer: The cart will have large labels; it will also include a clipboard for suggesting new titles. - Explanation: A semicolon correctly joins the two related independent clauses.

TACHS-WE7-003 - Correct: B - Correct Answer: Each Friday, volunteers will sort returns and deliver them to the appropriate rooms. - Explanation: Choice B clearly states when the task occurs and what volunteers will do without repetition.

TACHS-WE7-004 - Correct: D - Correct Answer: Sentence 7 - Explanation: The material of the front doors does not develop the book-cart proposal.

TACHS-WE7-005 - Correct: B - Correct Answer: The dots make the glass more visible to approaching birds while preserving light indoors. - Explanation: The plural subject "dots" requires the plural verb "make."

TACHS-WE7-006 - Correct: B - Correct Answer: During the following two weeks, we continued the same morning checks and recorded fewer incidents, although more observation is needed before a firm conclusion. - Explanation: Choice B combines the result and limitation in a complete, logically subordinated sentence.

TACHS-WE7-007 - Correct: D - Correct Answer: the location of each suspected collision - Explanation: The replacement is concise and precisely describes the recorded information.

TACHS-WE7-008 - Correct: D - Correct Answer: These windows became the focus of our first test. - Explanation: The sentence logically connects the observed collision pattern to the decision to treat those windows.

TACHS-WE7-009 - Correct: C - Correct Answer: The pair of gloves is in the top drawer. - Explanation: The singular subject "pair" correctly takes the verb "is."

TACHS-WE7-010 - Correct: A - Correct Answer: The internship required arriving early, taking careful notes, and asking questions. - Explanation: Choice A uses three parallel gerund phrases.

TACHS-WE7-011 - Correct: B - Correct Answer: Please divide the four notebooks among the students. - Explanation: Choice B clearly uses "among" for distribution within a group of more than two students.

TACHS-WE7-012 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 requires the preposition “to,” not the word “too.”

TACHS-WE7-013 - Correct: B - Correct Answer: Line 4 - Explanation: Line 4 uses a double comparative; “farther” should appear without “more.”

MATHEMATICS ANSWERS

TACHS-M7-001 - Correct: B - Correct Answer: $\frac{31}{36}$ - Explanation: Using denominator 36 gives $\frac{21}{36} + \frac{10}{36} = \frac{31}{36}$.

TACHS-M7-002 - Correct: A - Correct Answer: 7 - Explanation: Adding 11 gives $6x = 42$, so $x = 7$.

TACHS-M7-003 - Correct: C - Correct Answer: 15 - Explanation: 15 occurs twice, while every other value occurs once.

TACHS-M7-004 - Correct: D - Correct Answer: 68° - Explanation: The base angles share $180 - 44 = 136^\circ$, so each is 68° .

TACHS-M7-005 - Correct: C - Correct Answer: 6 - Explanation: 1.8 liters is 1,800 milliliters, and $1,800 \div 300 = 6$.

TACHS-M7-006 - Correct: D - Correct Answer: 5,472 - Explanation: The digits of 5,472 total 18, a multiple of 9.

TACHS-M7-007 - Correct: B - Correct Answer: Multiply by 2, then add 2. - Explanation: Each term equals twice the preceding term plus 2.

TACHS-M7-008 - Correct: D - Correct Answer: $\frac{1}{2}$ - Explanation: There are 6 favorable tiles out of 12, giving $\frac{1}{2}$.

TACHS-M7-009 - Correct: A - Correct Answer: 40 units - Explanation: The length is $96 \div 8 = 12$, so perimeter = $2(12 + 8) = 40$.

TACHS-M7-010 - Correct: B - Correct Answer: 260 miles - Explanation: The rate is 52 miles per hour, and $52 \times 5 = 260$ miles.

TACHS-M7-011 - Correct: A - Correct Answer: \$168 - Explanation: The increase is \$18, so the new price is \$168.

TACHS-M7-012 - Correct: A - Correct Answer: $\frac{p}{4} + 5$ - Explanation: The quotient is $\frac{p}{4}$, and five more gives $\frac{p}{4} + 5$.

ABILITY ANSWERS

TACHS-A7-001 - Correct: D - Correct Answer:   - Explanation: The filled square moves from left of the open square to its right.

TACHS-A7-002 - Correct: D - Correct Answer: ||| - Explanation: The lower cell has one additional vertical line, so two lines become three.

TACHS-A7-003 - Correct: A - Correct Answer: $\triangle + +$ - Explanation: The rule produces $\triangle$ followed by two plus signs: $\triangle + +$.

TACHS-A7-004 - Correct: A - Correct Answer: $\leftarrow \times$ - Explanation: The arrow opposite $\rightarrow$ is $\leftarrow$, and $+$ changes to $\times$.

TACHS-A7-005 - Correct: B - Correct Answer: $\{+, \times\}$ - Explanation: The shared symbols in the two Row 2 groups are $+$ and $\times$.

TACHS-A7-006 - Correct: A - Correct Answer: Two holes in a horizontal pair near the bottom - Explanation: The vertical unfold reflects the punch horizontally, creating two holes near the bottom.

TACHS-A7-007 - Correct: B - Correct Answer: Four holes forming two vertical pairs - Explanation: Each punch reflects across the horizontal fold, giving two vertical pairs and four holes.

TACHS-A7-008 - Correct: D - Correct Answer: Four quarter-circle cuts, one at each outer corner - Explanation: That folded corner stacks the four original outer corners, so unfolding places a quarter-circle cut at each corner.

TACHS-A7-009 - Correct: C - Correct Answer: One hole on the diagonal - Explanation: A punch on the fold line maps onto itself, so only one hole appears.

TACHS-A7-010 - Correct: D - Correct Answer:    - Explanation: Each group has one open shape followed by two filled versions of the same shape.

TACHS-A7-011 - Correct: D - Correct Answer: Top ; middle $+$; bottom $\triangle$ - Explanation: Two different open shapes appear above and below a central plus sign.

TACHS-A7-012 - Correct: C - Correct Answer: \ becomes / - Explanation: Each line is rotated 90°; a backslash becomes a forward slash.

Exam Part B

Filename: tachs_full_exam_07_part_b.txt

Exam Family: TACHS

Exam Number: 7

Exam Part: B

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R7-S04

Section: Reading

Stimulus Type: Literary Passage

Title: A Bell with Two Voices

Text:

In the back room of her aunt's bicycle shop, Tessa found a brass bell with a dull surface and a stiff lever. Her aunt explained that it had come from an old delivery bicycle and could be displayed in the front window if Tessa cleaned it.

Tessa polished the outside until it shone, but the bell produced only a weak click. She nearly placed it on the display shelf anyway. Then she removed the top and found two thin metal washers beneath the striker. One was bent, causing the striker to touch the bell before the lever was fully released.

The shop had no washer of the same size. Tessa flattened the bent one, reassembled the bell, and tested it. The ring returned, but a faint buzz followed each note. Rather than tightening every screw, she loosened the central screw by a quarter turn. The bell now had room to vibrate freely.

When her aunt heard it, she suggested mounting the bell where customers could try it. Tessa had expected the repaired object to become decoration. Instead, its clear ring drew people toward the window. The bell's appearance told visitors that it was old; its sound showed them that age and usefulness did not have to be opposites.

STIMULUS END

TACHS-R7-014

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S04

Prompt:

What causes the bell's first weak click?

- A) A bent washer holds the striker in the wrong position.
- B) The brass surface has not been polished.
- C) The central screw is completely missing.
- D) The lever is attached to the display shelf.

TACHS-R7-015

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S04

Prompt:

Why does loosening the central screw remove the buzz?

- A) It changes the bell's color.
- B) It prevents customers from touching the lever.
- C) It gives the metal enough freedom to vibrate.
- D) It bends the second washer to match the first.

TACHS-R7-016

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S04

Prompt:

Which statement best expresses a central idea of the passage?

- A) Objects are valuable only when they look new.
- B) Old tools should remain untouched in displays.
- C) A repair succeeds only when every part is replaced.
- D) Careful adjustment can restore both function and appreciation to an old object.

TACHS-R7-017

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S04

Prompt:

The contrast between how the bell looks and how it sounds mainly emphasizes

- A) why brass should never be polished.
- B) that customers prefer new bicycles.
- C) that the shop window blocks sound.
- D) that visible age can coexist with continued usefulness.

STIMULUS START

Stimulus ID: TACHS-R7-S05

Section: Reading

Stimulus Type: Scientific Passage

Title: When Tree Roots Connect

Text:

The roots of neighboring trees usually grow as separate systems, but roots of the same or closely related species sometimes press together and join. This natural connection is called a root graft. Over time, tissues at the contact point may fuse, creating a pathway between the trees.

Through a graft, water, minerals, or stored sugars may move from one root system to another. A shaded tree with limited access to sunlight might receive some resources from a healthier neighbor. Connections can also help a stump remain alive after its trunk has been cut, because attached trees continue supplying its roots.

The connection is not always beneficial. A disease-causing organism can sometimes travel through linked roots. In managed forests or orchards, understanding which trees are connected may

therefore matter when workers try to limit disease spread.

Scientists study grafts by tracing substances placed in one tree or by examining root tissue. The effects vary with species, distance, soil, and the condition of each tree. A root graft does not mean an entire forest functions as one organism, but it does show that boundaries between neighboring trees may be less complete beneath the soil than they appear above it.

STIMULUS END

TACHS-R7-018

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S05

Prompt:

What is a root graft?

- A) A container used to grow seedlings
- B) A disease found only in orchard soil
- C) A fused connection between neighboring root systems
- D) A cut made to measure a tree's age

TACHS-R7-019

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S05

Prompt:

As used in the final paragraph, "tracing" most nearly means

- A) guessing the color of.
- B) drawing an outline around.
- C) removing all evidence of.
- D) following the movement of.

TACHS-R7-020

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R7-S05

Prompt:

How is the passage mainly organized?

- A) It describes a discovery in a single tree from beginning to end.
- B) It compares tree heights in several forests.
- C) It defines root grafts, explains possible effects, and describes how they are studied.
- D) It argues that every tree in a forest shares one root system.

TACHS-R7-021

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S05

Prompt:

Why might orchard workers remove more than one tree near a diseased tree?

- A) Healthy roots always stop producing stored sugars.
- B) Linked roots could allow the disease organism to spread.
- C) Removing trees guarantees that soil becomes deeper.
- D) Every nearby tree must be the same age.

STIMULUS START

Stimulus ID: TACHS-R7-S06

Section: Reading

Stimulus Type: Historical-Style Passage

Title: Keeping the Lighthouse Lens Turning

Text:

In many nineteenth-century lighthouses, a large lens rotated around a lamp to create a repeating flash pattern. Sailors could identify a lighthouse by the timing of its flashes. Producing that pattern required more than a bright flame; the heavy lens had to turn at a steady speed.

Some rotation systems used a descending weight, much like a tall clock. As gravity pulled the weight downward, gears turned the lens. A keeper regularly wound the weight back to the top before it reached the bottom. If the mechanism slowed because of dirt or worn parts, the flash interval could change.

Keepers cleaned soot from glass, trimmed lamp wicks, lubricated gears, and recorded observations in daily logs. They also watched weather conditions because condensation and salt could affect equipment. Routine work prevented small problems from becoming failures during storms, when the light was especially important.

Modern lighthouses often use electric lamps and automated controls, but the older systems reveal an important principle. A signal may appear simple to a distant observer while depending on precise machinery and repeated maintenance. The familiar flash across the water was not only a product of the lens. It was also the result of work performed between flashes.

STIMULUS END

TACHS-R7-022

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S06

Prompt:

What powered some lighthouse lens rotation systems?

- A) A descending weight
- B) Ocean waves striking the tower
- C) Heat rising from the lamp
- D) Sailors pulling an exterior rope

TACHS-R7-023

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S06

Prompt:

As used in the second paragraph, “interval” most nearly means

- A) brightness.
- B) period of time between events.
- C) distance from shore.
- D) amount of oil in a lamp.

TACHS-R7-024

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S06

Prompt:

Which statement best summarizes the passage?

- A) Lighthouse lenses rotated without human attention.
- B) Sailors repaired lighthouse gears from their ships.
- C) Electric lamps produced the first recognizable signals.
- D) Reliable flash patterns depended on precise mechanisms and regular maintenance.

TACHS-R7-025

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S06

Prompt:

Why does the author end with “work performed between flashes”?

- A) To suggest that keepers worked only in darkness
- B) To claim that maintenance made the lens less useful
- C) To explain that sailors could see the keepers through the lens
- D) To highlight the unseen effort behind a simple visible signal.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE7-S03

Section: Written Expression

Stimulus Type: Revision Passage

Title: Improving the Lost-and-Found

Text:

[1] Our school’s lost-and-found bin fills quickly, and many items are never claimed. [2] The student council proposes adding numbered tags to items as they arrive. [3] A weekly online list would describe each item without revealing every identifying detail. [4] A student claiming a jacket, for example, might be asked to name the brand or an object left in its pocket. [5] The bin is beside a wall painted pale gray. [6] Items would remain available for six weeks before being donated according to school policy. [7] This system could make searches faster while protecting items from incorrect

claims. [8] It would also provide a clear record of how long each object has been stored.

STIMULUS END

TACHS-WE7-014

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S03

Prompt:

Which sentence should be removed because it is irrelevant?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 5
- D) Sentence 8

TACHS-WE7-015

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S03

Prompt:

Which revision of sentence 3 is most precise?

- A) A thing online would say some information about lost stuff.
- B) Each Friday, a password-protected page would list general descriptions and tag numbers for newly found items.
- C) Every week, the Internet list will have items and descriptions and all that.
- D) A list would be made in a way that does not tell everything.

TACHS-WE7-016

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S03

Prompt:

Where would the following sentence fit best? "The tag number would also appear on the item's online listing."

- A) Before sentence 1
- B) After sentence 6
- C) After sentence 2
- D) After sentence 8

TACHS-WE7-017

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S03

Prompt:

Which revision most effectively combines sentences 7 and 8?

- A) By speeding searches, discouraging incorrect claims, and recording storage time, this system would improve management of found items.
- B) Protecting items and it would make searches fast, recording time.
- C) The system could be faster while claims, also records that are clear.
- D) This system searches faster, incorrect claims, and a clear storage record.

STIMULUS START

Stimulus ID: TACHS-WE7-S04

Section: Written Expression

Stimulus Type: Revision Passage

Title: Measuring Cafeteria Waste

Text:

[1] The cafeteria committee wanted to learn which foods were discarded most often. [2] For five lunches, volunteers placed labeled collection tubs beside the tray return. [3] Separate tubs held unopened packages, fruit scraps, and other leftovers. [4] At the end of lunch, each tub were weighed with the same scale. [5] Volunteers wore gloves and did not remove any material from the tubs. [6] The results showed that unopened milk cartons accounted for the greatest mass. [7] Before changing the menu, the committee plans to repeat the study during a different month. [8] A second sample may show whether the first pattern was typical.

STIMULUS END

TACHS-WE7-018

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE7-S04

Prompt:

Which revision corrects the error in sentence 4?

- A) At the end of lunch, each tub were weighed with the same scale.
- B) At the end of lunch, each tub was weighed with the same scale.
- C) At the end of lunch, each tubs was weighed with the same scale.
- D) At the end of lunch, each tub have been weighed with the same scale.

TACHS-WE7-019

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S04

Prompt:

Which phrase best replaces “accounted for the greatest mass” in sentence 6?

- A) were a lot of the stuff by weight
- B) were the heaviest category collected
- C) made up the biggest kind of thing
- D) had a mass that was greatly the greatest

TACHS-WE7-020

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S04

Prompt:

Which sentence best explains why the committee will repeat the study?

- A) The same scale is difficult to carry.
- B) Volunteers want to use larger tubs next time.
- C) A sample from another time can show whether the result is consistent.
- D) Fruit scraps always weigh less in winter.

TACHS-WE7-021

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S04

Prompt:

Which revision most effectively combines sentences 7 and 8?

- A) Before changing the menu and another month, a repeated study, whether typical.
- B) Before changing the menu, the committee will repeat the study in another month to determine whether the first pattern was typical.
- C) The committee plans another month, changing the menu after a pattern.
- D) A second sample may be typical because the menu will change and repeat.

TACHS-WE7-022

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) There are three reasons to revise the plan.
- B) There is three reasons to revise the plan.
- C) There was three reasons to revise the plan.
- D) There be three reasons to revise the plan.

TACHS-WE7-023

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) She had drank the water before the race.
- B) She has drink the water before the race.
- C) She had drunk the water before the race.
- D) She has drank the water before the race.

TACHS-WE7-024

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The audience waited quietly for the lecture.

Line 2: Two speakers arranged notes on the table.

Line 3: The first presentation began at noon.

Line 4: The speaker gave we a printed outline.

- A) Line 4
- B) Line 2
- C) Line 3
- D) Line 1

TACHS-WE7-025

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: We compared the two fabric samples.

Line 2: The first felt smoother then the second.

Line 3: Both samples resisted water well.

Line 4: Our notes included the results of each test.

- A) Line 2
- B) Line 1
- C) Line 3
- D) Line 4

TACHS-WE7-026

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The hikers reached the shelter before sunset.

Line 2: They hung wet jackets near the entrance.

Line 3: A small stove did not warm no part of the room.

Line 4: Everyone prepared a simple evening meal.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M7-013

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the value of $4^2 + 3(7 - 5)$?

A) 28

B) 24

C) 22

D) 38

TACHS-M7-014

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The mean of 7, 9, 12, and x is 10. What is x ?

A) 10

B) 16

C) 14

D) 12

TACHS-M7-015

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A regular octagon has a perimeter of 72 inches. How long is each side?

A) 12 inches

- B) 8 inches
- C) 10 inches
- D) 9 inches

TACHS-M7-016

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A concert starts at 7:25 p.m. and ends at 9:08 p.m. How long does it last?

- A) 1 hour 33 minutes
- B) 2 hours 17 minutes
- C) 1 hour 43 minutes
- D) 2 hours 43 minutes

TACHS-M7-017

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which value is least?

- A) -3.2
- B) -3.08
- C) $-3 \frac{1}{4}$
- D) $-3 \frac{1}{5}$

TACHS-M7-018

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $5(x + 1) = 40$.

- A) 5
- B) 7
- C) 8
- D) 9

TACHS-M7-019

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A number cube is rolled twice. What is the probability of rolling a 6 both times?

- A) $\frac{1}{6}$
- B) $\frac{1}{12}$
- C) $\frac{1}{36}$
- D) $\frac{1}{72}$

TACHS-M7-020

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A cylinder has radius 3 cm and height 5 cm. Using $\pi = 3.14$, what is its volume?

- A) 141.3 cm^3
- B) 94.2 cm^3
- C) 47.1 cm^3
- D) 282.6 cm^3

TACHS-M7-021

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The data are 3, 5, 5, 7, 10. If 15 is added, which measure increases the most?

- A) Mode
- B) Median
- C) Minimum
- D) Mean

TACHS-M7-022

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A car uses 9 gallons to travel 243 miles. At the same rate, how far can it travel on 14 gallons?

- A) 324 miles
- B) 351 miles
- C) 392 miles
- D) 378 miles

TACHS-M7-023

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is the least common multiple of 8 and 14?

- A) 28
- B) 42
- C) 56
- D) 48

TACHS-M7-024

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

If $g(x) = 4x + 1$, which value of x gives $g(x) = 29$?

- A) 7
- B) 5
- C) 6
- D) 8

TACHS-M7-025

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A triangle has base 16 meters and area 56 square meters. What is its height?

- A) 7 meters
- B) 3.5 meters
- C) 14 meters
- D) 28 meters

SECTION 4 - ABILITY**TACHS-A7-013**

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the last symbol moves to the front.

Row 1: ○ □ △ | △ ○ □

Row 2: + × ★ | ?

A) × ★ +

B) ★ + ×

C) + ★ ×

D) ★ × +

TACHS-A7-014

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, two open shapes become one filled shape of the same kind.

Row 1: ○ ○ | ●

Row 2: □ □ | ?

A) □

B) ●

C) ■ ■

D) ■

TACHS-A7-015

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, the symbol inside and the symbol outside exchange positions.

Top-left: + inside ○

Bottom-left: ○ inside +

Top-right: × inside □

Bottom-right: ?

A) × inside □

B) ■ inside ×

C) × beside □

D) □ inside ×

TACHS-A7-016

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. From left to right, the arrow turns 90° counterclockwise and one + is added.

Cell 1: → +

Cell 2: ↑ + +

Cell 3: ?

A) → + + +

B) ← + + +

C) ↑ + + +

D) ↓ + + +

TACHS-A7-017

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from top to bottom. A hole is punched near the folded paper's right edge and a second hole near its left edge, at different distances from the fold. What appears after unfolding?

A) Eight holes at equal distances from the center

B) Two holes near the bottom edge

C) Four holes in one horizontal line

D) Four holes forming two vertical mirrored pairs

TACHS-A7-018

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded from left to right. A semicircular cut is made on the vertical folded edge, away from top and bottom. What appears after unfolding?

A) One complete circular opening centered on the vertical centerline

B) Two semicircles on the outer edges

C) Four circular openings

D) One semicircle on the right edge

TACHS-A7-019

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded top to bottom and then left to right. A hole is punched near the center of the folded square, away from edges. What appears after unfolding?

- A) Two holes across the top
- B) Four holes on one diagonal
- C) Eight holes around the perimeter
- D) Four holes, one in each quadrant

TACHS-A7-020

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded along the diagonal from lower-left to upper-right. A notch is cut from the folded paper's lower outer edge, away from corners and the fold. What appears after unfolding?

- A) One notch at the center
- B) Two notches on adjacent outer edges, mirrored across the diagonal
- C) Four notches near all corners
- D) Two notches on the same lower edge

TACHS-A7-021

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following figures share one rule:

1. Outer ○; inner □; + below
2. Outer △; inner ◇; + below
3. Outer ☆; inner ○; + below

Which choice follows the same rule?

- A) Outer □; inner △; × below
- B) Outer ■; inner △; + below
- C) Outer □; inner △; + below
- D) Outer ○; inner ○; + below

TACHS-A7-022

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share one rule:

1. ● ○ ○
2. ■ □ □

3. ▲ △ △

Which choice follows the same rule?

A) ◇ ◆ ◆

B) ◆ ◇ ◇

C) ☆ ★ ★

D) ○ ● ○

TACHS-A7-023

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. Left ↑; center ○; right ↓

2. Left ←; center □; right →

3. Left ↓; center △; right ↑

Which choice follows the same rule?

A) Left →; center ◇; right ←

B) Left ↑; center ●; right ↓

C) Left →; center ■; right ←

D) Left →; center ◇; right ↑

TACHS-A7-024

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following sequences share one rule:

1. ○ + □ ×

2. △ + ◇ ×

3. ☆ + ○ ×

Which choice follows the same rule?

A) □ + △ ×

B) ■ + □ ×

C) □ + □ ×

D) □ × △ +

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R7-014 - Correct: A - Correct Answer: A bent washer holds the striker in the wrong position. - Explanation: The bent washer makes the striker touch the bell before the lever is fully released.

TACHS-R7-015 - Correct: C - Correct Answer: It gives the metal enough freedom to vibrate. - Explanation: The passage explains that slight loosening allows the bell to vibrate freely.

TACHS-R7-016 - Correct: D - Correct Answer: Careful adjustment can restore both function and appreciation to an old object. - Explanation: Tessa's careful investigation restores the bell's sound and turns it into an engaging display.

TACHS-R7-017 - Correct: D - Correct Answer: that visible age can coexist with continued usefulness. - Explanation: The bell looks old yet rings clearly, reinforcing the final idea that age does not prevent usefulness.

TACHS-R7-018 - Correct: C - Correct Answer: A fused connection between neighboring root systems - Explanation: A root graft forms when contacting roots fuse and create a pathway between trees.

TACHS-R7-019 - Correct: D - Correct Answer: following the movement of. - Explanation: Researchers follow where a placed substance moves to learn whether roots are connected.

TACHS-R7-020 - Correct: C - Correct Answer: It defines root grafts, explains possible effects, and describes how they are studied. - Explanation: The passage proceeds from definition to benefits and risks, then methods and limits of study.

TACHS-R7-021 - Correct: B - Correct Answer: Linked roots could allow the disease organism to spread. - Explanation: Because pathogens may travel through grafted roots, connected neighboring trees could also be at risk.

TACHS-R7-022 - Correct: A - Correct Answer: A descending weight - Explanation: The passage compares the descending weight system with the mechanism of a tall clock.

TACHS-R7-023 - Correct: B - Correct Answer: period of time between events. - Explanation: The flash interval is the period of time between repeated flashes.

TACHS-R7-024 - Correct: D - Correct Answer: Reliable flash patterns depended on precise mechanisms and regular maintenance. - Explanation: The passage emphasizes the machinery and keeper tasks required to maintain a dependable signal.

TACHS-R7-025 - Correct: D - Correct Answer: To highlight the unseen effort behind a simple visible signal. - Explanation: The phrase emphasizes that each reliable flash depended on ongoing maintenance not visible from sea.

WRITTEN EXPRESSION ANSWERS

TACHS-WE7-014 - Correct: C - Correct Answer: Sentence 5 - Explanation: The wall color does not explain the tagging, claiming, or storage procedure.

TACHS-WE7-015 - Correct: B - Correct Answer: Each Friday, a password-protected page would list general descriptions and tag numbers for newly found items. - Explanation: Choice B specifies timing, access, content, and the privacy-protecting level of detail.

TACHS-WE7-016 - Correct: C - Correct Answer: After sentence 2 - Explanation: After sentence 2, the added sentence directly connects the physical tag with the proposed online list.

TACHS-WE7-017 - Correct: A - Correct Answer: By speeding searches, discouraging incorrect claims, and recording storage time, this system would improve management of found items. - Explanation: Choice A combines all three benefits in a complete sentence with parallel structure.

TACHS-WE7-018 - Correct: B - Correct Answer: At the end of lunch, each tub was weighed with the same scale. - Explanation: The singular subject "each tub" requires the singular past-tense verb "was."

TACHS-WE7-019 - Correct: B - Correct Answer: were the heaviest category collected - Explanation: The revision expresses the result directly and clearly in age-appropriate formal language.

TACHS-WE7-020 - Correct: C - Correct Answer: A sample from another time can show whether the result is consistent. - Explanation: Repeating the study at another time tests whether the first pattern is stable rather than unusual.

TACHS-WE7-021 - Correct: B - Correct Answer: Before changing the menu, the committee will repeat the study in another month to determine whether the first pattern was typical. - Explanation: Choice B clearly connects the repeated study with its purpose and the decision it will inform.

TACHS-WE7-022 - Correct: A - Correct Answer: There are three reasons to revise the plan. - Explanation: The plural noun "reasons" requires "There are."

TACHS-WE7-023 - Correct: C - Correct Answer: She had drunk the water before the race. - Explanation: The past participle of "drink" is "drunk," correctly used after "had."

TACHS-WE7-024 - Correct: A - Correct Answer: Line 4 - Explanation: Line 4 requires the object pronoun "us" after the verb "gave."

TACHS-WE7-025 - Correct: A - Correct Answer: Line 2 - Explanation: Line 2 requires "than," the word used in comparisons, rather than "then."

TACHS-WE7-026 - Correct: C - Correct Answer: Line 3 - Explanation: Line 3 uses a nonstandard double negative; "did not warm any part" is correct.

MATHEMATICS ANSWERS

TACHS-M7-013 - Correct: C - Correct Answer: 22 - Explanation: $4^2 + 3(2) = 16 + 6 = 22$.

TACHS-M7-014 - Correct: D - Correct Answer: 12 - Explanation: A mean of 10 requires total 40; $40 - 28 = 12$.

TACHS-M7-015 - Correct: D - Correct Answer: 9 inches - Explanation: A regular octagon has 8 equal sides, so $72 \div 8 = 9$ inches.

TACHS-M7-016 - Correct: C - Correct Answer: 1 hour 43 minutes - Explanation: From 7:25 to 8:00 is 35 minutes, then 68 minutes more, totaling 103 minutes or 1 hour 43 minutes.

TACHS-M7-017 - Correct: C - Correct Answer: $-3 \frac{1}{4}$ - Explanation: $-3 \frac{1}{4} = -3.25$, which is less than -3.2 , -3.08 , and -3.2 .

TACHS-M7-018 - Correct: B - Correct Answer: 7 - Explanation: Divide by 5 to get $x + 1 = 8$, so $x = 7$.

TACHS-M7-019 - Correct: C - Correct Answer: $\frac{1}{36}$ - Explanation: The independent probability is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

TACHS-M7-020 - Correct: A - Correct Answer: 141.3 cm^3 - Explanation: Volume = $\pi r^2 h = 3.14 \times 9 \times 5 = 141.3$ cubic centimeters.

TACHS-M7-021 - Correct: D - Correct Answer: Mean - Explanation: The high added value raises the mean from 6 to 7.5, while the median rises only from 5 to 6.

TACHS-M7-022 - Correct: D - Correct Answer: 378 miles - Explanation: The rate is $243 \div 9 = 27$ miles per gallon; $27 \times 14 = 378$ miles.

TACHS-M7-023 - Correct: C - Correct Answer: 56 - Explanation: $8 = 2^3$ and $14 = 2 \times 7$, so the LCM is $2^3 \times 7 = 56$.

TACHS-M7-024 - Correct: A - Correct Answer: 7 - Explanation: Solve $4x + 1 = 29$: $4x = 28$, so $x = 7$.

TACHS-M7-025 - Correct: A - Correct Answer: 7 meters - Explanation: $56 = \frac{1}{2} \times 16 \times h = 8h$, so $h = 7$ meters.

ABILITY ANSWERS

TACHS-A7-013 - Correct: B - Correct Answer: $\star + \times$ - Explanation: Moving the last symbol $\star$ to the front gives $\star + \times$.

TACHS-A7-014 - Correct: D - Correct Answer: ■ - Explanation: Two open squares combine into one filled square.

TACHS-A7-015 - Correct: D - Correct Answer: □ inside $\times$ - Explanation: Exchanging positions puts □ inside $\times$.

TACHS-A7-016 - Correct: B - Correct Answer: $\leftarrow + + +$ - Explanation: Turning $\uparrow$ another 90° counterclockwise gives $\leftarrow$, and a third plus is added.

TACHS-A7-017 - Correct: D - Correct Answer: Four holes forming two vertical mirrored pairs - Explanation: Each of the two punches reflects vertically, producing two mirrored pairs and four holes.

TACHS-A7-018 - Correct: A - Correct Answer: One complete circular opening centered on the vertical centerline - Explanation: The semicircle on the fold reflects to complete one circle centered on the fold line.

TACHS-A7-019 - Correct: D - Correct Answer: Four holes, one in each quadrant - Explanation: Two perpendicular folds reflect the punch into four corresponding quadrant positions.

TACHS-A7-020 - Correct: B - Correct Answer: Two notches on adjacent outer edges, mirrored across the diagonal - Explanation: Reflecting the outer-edge notch across the diagonal places its mate on the adjacent outer edge.

TACHS-A7-021 - Correct: C - Correct Answer: Outer □; inner △; + below - Explanation: One open shape contains a different open shape, with a plus sign below.

TACHS-A7-022 - Correct: B - Correct Answer: ◆ ◇ ◇ - Explanation: Each group begins with one filled shape followed by two matching open versions.

TACHS-A7-023 - Correct: A - Correct Answer: Left →; center ◇; right ← - Explanation: Opposite arrows surround one open shape; → and ← are opposite around open ◇.

TACHS-A7-024 - Correct: A - Correct Answer: □ + △ × - Explanation: Two different open shapes occupy positions 1 and 3, with + second and × fourth.

Exam Part C

Filename: tachs_full_exam_07_part_c.txt

Exam Family: TACHS

Exam Number: 7

Exam Part: C

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R7-S07

Section: Reading

Stimulus Type: Literary Passage

Title: The Photograph That Waited

Text:

For the photography club exhibit, Omar built a pinhole camera from a small wooden box. Light entered through a tiny opening and formed an image on photographic paper inside. His first picture of the courtyard was pale because he had uncovered the pinhole for only a few seconds.

On his second attempt, Omar chose a row of bicycles and exposed the paper for two minutes. Halfway through, a custodian moved one bicycle. Omar groaned, certain the picture was ruined. When he developed the paper, however, the moved bicycle appeared as a faint, transparent shape beside its darker final position.

Omar planned to repeat the photograph, but he kept studying the ghostlike bicycle. The camera had not frozen one instant. It had collected light throughout the exposure, recording both stillness and change. He returned to the courtyard during dismissal and made a longer exposure. The building and bike rack appeared solid, while passing students became soft streaks.

At the exhibit, Omar displayed the two images together under the title "What Stayed, What Passed." His first successful photograph was not the perfectly sharp scene he had intended. It was the one that taught him what his camera could notice.

STIMULUS END

TACHS-R7-026

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S07

Prompt:

Why is Omar's first courtyard picture pale?

- A) The pinhole was uncovered for too little time.
- B) The wooden box allowed too much light inside.
- C) A custodian moved the camera.
- D) The photographic paper was already developed.

TACHS-R7-027

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S07

Prompt:

Why does the moved bicycle appear faint in its first position?

- A) Its paint reflects no light.
- B) It remained there for only part of the exposure.
- C) Omar erased it during development.
- D) The pinhole formed two separate photographs.

TACHS-R7-028

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S07

Prompt:

Which statement best expresses a central idea of the passage?

- A) Every photograph should capture a perfectly still instant.
- B) An unexpected result can reveal a tool's distinctive possibilities.
- C) Pinhole cameras work only when objects move.
- D) A successful exhibit requires many identical photographs.

TACHS-R7-029

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S07

Prompt:

As used in the third paragraph, "frozen" most nearly means

- A) captured as if time had stopped.
- B) covered with ice.
- C) unable to be developed.
- D) made extremely cold.

STIMULUS START

Stimulus ID: TACHS-R7-S08

Section: Reading

Stimulus Type: Literary Passage

Title: The Initials Under the Paint

Text:

Sofia was helping refinish wooden desks for the youth center. Most had layers of chipped paint, so volunteers sanded them smooth before applying a new finish. On the underside of one desktop, Sofia found several initials and dates carved lightly into the wood.

The project leader told her the marks could be sanded away. Sofia wondered who had made them and why some dates were decades apart. She photographed the carvings, then examined the desk more closely. The top was damaged beyond repair, but the underside remained strong.

Instead of removing the marks, Sofia suggested turning the desktop over. The carved side could not become the new writing surface because the grooves would catch pencils. The team therefore cut a

narrow section from the marked wood, sealed it, and attached it beneath the desk's front edge. The remaining wood was used for braces.

When the desk returned to the center, its new top was smooth, but the strip of initials remained visible to anyone seated there. Sofia added no explanation beside it. She liked that the marks did not tell a complete story. They simply showed that the desk's new users were not its first.

STIMULUS END

TACHS-R7-030

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S08

Prompt:

Where does Sofia first find the carved initials?

- A) On the desk's new writing surface
- B) Inside a wooden brace
- C) On the underside of the old desktop
- D) Beneath the youth center floor

TACHS-R7-031

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S08

Prompt:

Why does the team seal the marked strip of wood?

- A) To make the carved grooves deeper
- B) To preserve it as part of the rebuilt desk
- C) To prepare it for use as a pencil
- D) To hide it behind the wall

TACHS-R7-032

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S08

Prompt:

Why does Sofia prefer not to add an explanation beside the initials?

- A) She knows exactly who carved every mark.
- B) She wants visitors to believe the desk is new.
- C) She plans to sand the marks away later.
- D) She values the marks as evidence of earlier users even without a complete story.

TACHS-R7-033

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R7-S08

Prompt:

How is the central problem resolved?

- A) The damaged desk is discarded and replaced.
- B) A marked section is preserved while the desk is rebuilt for practical use.
- C) The carved side becomes the new writing surface.
- D) The initials are copied onto the wall.

STIMULUS START

Stimulus ID: TACHS-R7-S09

Section: Reading

Stimulus Type: Informational Passage

Title: A Tide Pool's Daily Swing

Text:

A tide pool may look like a small, stable pond, but its conditions can change dramatically within hours. At high tide, ocean water flows over the pool, mixing its contents with cooler water rich in dissolved oxygen. At low tide, the pool becomes isolated.

Sunlight can warm shallow water quickly. Meanwhile, algae and other organisms perform photosynthesis, releasing oxygen during daylight. After sunset, photosynthesis stops, but plants, animals, and microbes continue using oxygen through respiration. As a result, oxygen levels may fall during the night and reach their lowest point before morning.

Salinity also changes. Evaporation can leave salts behind and make the pool saltier, while rainfall can dilute it. A deep pool shaded by rock may change less rapidly than a shallow pool exposed to direct sun and wind.

Organisms living there must tolerate variation or find protected spaces. Some close their shells, hide beneath rocks, or reduce activity during stressful periods. Researchers who measure a tide pool only once may miss these cycles. Repeated measurements across a full day provide a more accurate picture of the environment its residents actually experience.

STIMULUS END

TACHS-R7-034

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S09

Prompt:

When may oxygen in a tide pool reach its lowest level?

- A) During every rainfall
- B) Immediately after high tide
- C) At the brightest point of midday
- D) Just before morning

TACHS-R7-035

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S09

Prompt:

As used in the third paragraph, “dilute” most nearly means

- A) make less concentrated.
- B) heat rapidly.
- C) remove all organisms from.
- D) increase the salt content of.

TACHS-R7-036

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S09

Prompt:

Which statement best summarizes the passage?

- A) Tide-pool conditions vary with tides, light, weather, and time, requiring adaptation and repeated study.
- B) All tide-pool organisms respond to stress in the same way.
- C) Tide pools remain constant whenever they are isolated.
- D) Deep pools always contain less oxygen than shallow pools.

TACHS-R7-037

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S09

Prompt:

Which pool would probably experience the fastest daytime warming?

- A) A deep pool shaded by a rock ledge
- B) A pool continuously connected to cool ocean water
- C) A shallow pool exposed to direct sunlight.
- D) A covered indoor saltwater tank

TACHS-R7-038

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S09

Prompt:

Why does the author discuss measurements across a full day?

- A) To show that a single observation may not represent changing conditions
- B) To prove that low tide lasts exactly twenty-four hours
- C) To argue that researchers should avoid tide pools at night
- D) To suggest that rainfall has no measurable effect

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE7-S05

Section: Written Expression

Stimulus Type: Revision Passage

Title: Auditing an Accessible Route

Text:

[1] Our student committee examined the route from the front gate to the auditorium. [2] We wanted to identify obstacles for visitors using wheelchairs, walkers, or strollers. [3] Committee members measured doorway widths, ramp slopes, and the force needed to open several doors. [4] We also walked the route during dismissal, when crowded hallways created different obstacles. [5] The east entrance had a power-assisted door, the button was partly hidden behind a planter. [6] A directional sign pointed toward an elevator that was temporarily out of service. [7] Our report recommends moving the planter, updating temporary signs, and repeating the route check each semester. [8] Accessibility depends not only on permanent construction but also on how a route is maintained and used.

STIMULUS END

TACHS-WE7-027

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S05

Prompt:

Which sentence best states the purpose of the committee's examination?

- A) Sentence 2
- B) Sentence 1
- C) Sentence 5
- D) Sentence 8

TACHS-WE7-028

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S05

Prompt:

Which revision corrects the error in sentence 5?

- A) The east entrance had a power-assisted door, however the button partly hidden behind a planter.
- B) The east entrance had a power-assisted door; however, the button was partly hidden behind a planter.
- C) The east entrance having a power-assisted door, the button behind a planter.
- D) Although the east entrance had a power-assisted door and the hidden button.

TACHS-WE7-029

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S05

Prompt:

Which revision of sentence 4 is most precise?

- A) We also did it again when lots of people were around and things changed.
- B) The route was walked by us at a different time with crowds and obstacles.
- C) We repeated the walk during dismissal to observe how hallway traffic affected the route.
- D) Dismissal happened, and we went there because it was another situation.

TACHS-WE7-030

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S05

Prompt:

Which sentence would best follow sentence 6?

- A) The elevator's interior walls are painted silver.
- B) Auditorium performances often begin at seven o'clock.
- C) Many students prefer using the stairs.
- D) Without a corrected sign, a visitor could travel to a route that cannot be completed.

STIMULUS START

Stimulus ID: TACHS-WE7-S06

Section: Written Expression

Stimulus Type: Revision Passage

Title: Testing Protective Packaging

Text:

[1] Our engineering class compared three ways to protect a fragile clay disk during a short drop. [2] Each disk was placed in the same size box. [3] One box contained crumpled paper, one contained folded cardboard strips, and one contained cotton batting. [4] We dropped every box from a height of one meter onto the same surface. [5] After each drop, the disk was remove and inspected for cracks. [6] The cardboard-strip package prevented cracks while using the least mass of material. [7] A future trial should repeat each packaging method several times. [8] More trials would show whether the first results are dependable rather than accidental.

STIMULUS END

TACHS-WE7-031

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE7-S06

Prompt:

Which revision corrects the error in sentence 5?

- A) After each drop, the disk was removing and inspected for cracks.
- B) After each drop, the disk were removed and inspected for cracks.
- C) After each drop, the disk has remove and inspected for cracks.
- D) After each drop, the disk was removed and inspected for cracks.

TACHS-WE7-032

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S06

Prompt:

Which revision most effectively combines sentences 7 and 8?

- A) Repeating each method several times would show whether the initial results are dependable rather than accidental.
- B) A future trial, each method several times, and dependable results rather than accidental.
- C) Because results accidental, repeating packages, several times in a future trial.
- D) More trials are future and should repeat because each method is dependable.

TACHS-WE7-033

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S06

Prompt:

Which phrase best replaces “using the least mass of material” in sentence 6?

- A) with the least massive stuff used up
- B) while using the lowest mass of packing material
- C) while material was the least kind by mass
- D) and it used material that did not weigh a lot

TACHS-WE7-034

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The equipment need to be cleaned after each use.
- B) The equipment have needed to be cleaned yesterday.
- C) The equipments need cleaning after every uses.
- D) The equipment needs to be cleaned after each use.

TACHS-WE7-035

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: None

Prompt:

Which choice is a complete sentence?

- A) While the orchestra tuned before the concert.
- B) Because the audience had already entered.
- C) Tuning before the concert in the crowded hall.
- D) The orchestra tuned while the audience entered.

TACHS-WE7-036

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: None

Prompt:

Which sentence is most concise?

- A) The reason we postponed the trip was because of the fact that rain was falling.
- B) Due to rain that was falling, that was why the trip was postponed by us.
- C) We postponed the trip because it was raining.
- D) The trip was postponed by us for the reason of rainy rain.

TACHS-WE7-037

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The twins chose separate topics for their reports.

Line 2: Ava researched the history of paper.

Line 3: Eli studied how ink was manufactured.

Line 4: Them presented their findings on Monday.

- A) Line 1
- B) Line 4
- C) Line 3
- D) Line 2

TACHS-WE7-038

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The ceramic bowl sat near the edge of the table.

Line 2: I moved it to a safer location.

Line 3: Its surface was smooth accept for one small ridge.

Line 4: The ridge had formed while the clay was drying.

- A) Line 3
- B) Line 2
- C) Line 1
- D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M7-026

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is $\frac{5}{6}$ of 42?

- A) 30
- B) 35
- C) 36
- D) 40

TACHS-M7-027

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $9 + 3x = 2x + 21$.

- A) 12
- B) 10
- C) 14
- D) 30

TACHS-M7-028

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The values are 11, 13, 16, 20, and 25. What is the mean?

- A) 15
- B) 17
- C) 16
- D) 18

TACHS-M7-029

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rhombus has diagonals of 10 cm and 16 cm. What is its area?

- A) 40 cm^2
- B) 52 cm^2
- C) 80 cm^2
- D) 160 cm^2

TACHS-M7-030

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A recipe calls for 450 grams of flour per loaf. How many kilograms are needed for 8 loaves?

- A) 2.6 kg
- B) 3.2 kg
- C) 3.6 kg
- D) 4.5 kg

TACHS-M7-031

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which number is prime?

- A) 51
- B) 57
- C) 63
- D) 59

TACHS-M7-032

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Which expression is equivalent to $4(2y - 3) + 5$?

- A) $6y + 2$

- B) $8y - 12$
- C) $8y - 7$
- D) $8y + 2$

TACHS-M7-033

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A bag contains 4 red, 5 blue, and 3 green marbles. What is the probability of drawing a marble that is not blue?

- A) $5/12$
- B) $7/12$
- C) $2/3$
- D) $3/4$

TACHS-M7-034

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

The angles of a quadrilateral measure 72° , 88° , 95° , and x° . What is x ?

- A) 105°
- B) 95°
- C) 100°
- D) 115°

TACHS-M7-035

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A printer produces 84 pages in 7 minutes. At the same rate, how many pages in 12 minutes?

- A) 124
- B) 132
- C) 156
- D) 144

TACHS-M7-036

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which is equivalent to 4.25?

A) $4 \frac{1}{5}$

B) $4 \frac{1}{4}$

C) $4 \frac{2}{5}$

D) $4 \frac{3}{4}$

TACHS-M7-037

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

If $h(t) = 2t^2 + 1$, what is $h(3)$?

A) 13

B) 17

C) 19

D) 37

SECTION 4 - ABILITY

TACHS-A7-025

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the first and third symbols exchange positions while the middle stays fixed.

Row 1: $\bigcirc + \square \mid \square + \bigcirc$

Row 2: $\triangle \times \diamond \mid ?$

A) $\diamond \triangle \times$

B) $\triangle \diamond \times$

C) $\times \triangle \diamond$

D) $\diamond \times \triangle$

TACHS-A7-026

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Down each column, the number of symbols doubles.

Top-left: ○

Bottom-left: ○ ○

Top-right: □ □

Bottom-right: ?

A) □ □ □

B) ■ ■ ■ ■

C) □ □ □ □

D) □ □ □ □ □ □

TACHS-A7-027

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell changes every filled shape to open and leaves positions unchanged.

Row 1: Top ●; bottom ■ | Top ○; bottom □

Row 2: Left ▲; right ◆ | ?

A) Left △; right ◆

B) Left △; right ◇

C) Left ▲; right ◇

D) Left ◇; right △

TACHS-A7-028

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. Across each row, the line rotates 90° clockwise and moves from left of the shape to right of it.

Row 1: | ○ | ○ -

Row 2: / □ | ?

A) \ □

B) □ /

C) □ \

D) / □

TACHS-A7-029

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the third cell contains all symbols from the first cell except those also in the second.

Row 1: {○, □, △} | {□} | {○, △}

Row 2: {+, ×, ★} | {×} | ?

- A) {+, ★}
- B) {×, ★}
- C) {+}
- D) {★}

TACHS-A7-030

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from bottom to top. A hole is punched near the folded paper's left outer edge, away from corners. What appears after unfolding?

- A) One hole at the center
- B) Two holes on the left side, mirrored across the horizontal centerline
- C) Two holes on opposite side edges
- D) Four holes near all corners

TACHS-A7-031

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded right to left and then bottom to top. A hole is punched near the folded paper's upper-left area, away from all edges. What appears after unfolding?

- A) Two holes in the upper half
- B) Four holes in one row
- C) Four holes in corresponding positions across both centerlines
- D) Eight holes near the perimeter

TACHS-A7-032

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded along the diagonal from upper-left to lower-right. A semicircular cut is made on the diagonal folded edge, away from corners. What appears after unfolding?

- A) Two separate semicircles
- B) Four complete circles
- C) One complete circular opening centered on the diagonal
- D) One semicircle on an outer edge

TACHS-A7-033

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from left to right. Two holes are punched on the vertical centerline of the folded rectangle at different heights, away from all edges. What appears after unfolding?

- A) Four holes forming two horizontal pairs
- B) Two holes remaining on one line
- C) Four holes forming two vertical pairs
- D) Eight holes in four rows

TACHS-A7-034

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. ○ inside □; × outside
2. △ inside ◇; × outside
3. ☆ inside ○; × outside

Which choice follows the same rule?

- A) ■ inside □; × outside
- B) ○ outside □; × inside
- C) □ inside △; + outside
- D) □ inside △; × outside

TACHS-A7-035

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following sequences share one rule:

1. ● ○ ■ □
2. ▲ △ ◆ ◇

3. ★ ☆ ● ○

Which choice follows the same rule?

A) ■ □ ▲ △

B) ◆ ◇ ▲ △

C) ○ ● □ ■

D) ■ ■ □ □

TACHS-A7-036

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

1. Top +; left ○; right □

2. Top +; left △; right ◇

3. Top +; left ☆; right ○

Which choice follows the same rule?

A) Top ×; left □; right △

B) Top +; left ■; right △

C) Top +; left □; right △

D) Top +; left □; right □

TACHS-A7-037

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following pairs share one rule:

1. ○ becomes ○ ○ ○

2. □ becomes □ □ □

3. △ becomes △ △ △

Which choice follows the same rule?

A) ◇ becomes ◇ ◇ ◇

B) ◆ becomes ◇ ◇ ◇

C) ☆ becomes ☆ ☆

D) + becomes + + + +

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R7-026 - Correct: A - Correct Answer: The pinhole was uncovered for too little time. - Explanation: The passage states that Omar exposed the first image for only a few seconds.

TACHS-R7-027 - Correct: B - Correct Answer: It remained there for only part of the exposure. - Explanation: The paper received light from the bicycle's first position for only part of the two-minute exposure.

TACHS-R7-028 - Correct: B - Correct Answer: An unexpected result can reveal a tool's distinctive possibilities. - Explanation: The unintended bicycle image teaches Omar how long exposure can record stillness and movement differently.

TACHS-R7-029 - Correct: A - Correct Answer: captured as if time had stopped. - Explanation: Omar realizes the exposure records a span of time rather than one motionless instant.

TACHS-R7-030 - Correct: C - Correct Answer: On the underside of the old desktop - Explanation: The initials and dates are carved lightly on the underside of the desktop.

TACHS-R7-031 - Correct: B - Correct Answer: To preserve it as part of the rebuilt desk - Explanation: Sealing the strip protects the carved marks before the piece is attached visibly to the desk.

TACHS-R7-032 - Correct: D - Correct Answer: She values the marks as evidence of earlier users even without a complete story. - Explanation: Sofia accepts that the initials offer partial evidence of the desk's history without identifying every person.

TACHS-R7-033 - Correct: B - Correct Answer: A marked section is preserved while the desk is rebuilt for practical use. - Explanation: The team saves the carved strip without using its uneven surface as the desktop.

TACHS-R7-034 - Correct: D - Correct Answer: Just before morning - Explanation: Respiration continues overnight without photosynthesis, so oxygen may be lowest before morning.

TACHS-R7-035 - Correct: A - Correct Answer: make less concentrated. - Explanation: Rainfall adds fresh water and makes the salt concentration lower.

TACHS-R7-036 - Correct: A - Correct Answer: Tide-pool conditions vary with tides, light, weather, and time, requiring adaptation and repeated study. - Explanation: The passage explains several daily changes, organism responses, and the need for measurements over time.

TACHS-R7-037 - Correct: C - Correct Answer: A shallow pool exposed to direct sunlight. - Explanation: Shallow water exposed to direct sun warms more rapidly than deep or shaded water.

TACHS-R7-038 - Correct: A - Correct Answer: To show that a single observation may not represent changing conditions - Explanation: Full-day measurements capture cycles that one isolated reading could miss.

WRITTEN EXPRESSION ANSWERS

TACHS-WE7-027 - Correct: A - Correct Answer: Sentence 2 - Explanation: Sentence 2 directly identifies the committee's goal and the visitors considered.

TACHS-WE7-028 - Correct: B - Correct Answer: The east entrance had a power-assisted door; however, the button was partly hidden behind a planter. - Explanation: A semicolon and conjunctive adverb correctly connect the contrasting independent clauses.

TACHS-WE7-029 - Correct: C - Correct Answer: We repeated the walk during dismissal to observe how hallway traffic affected the route. - Explanation: Choice C clearly states the repeated action, timing, and reason for observing the route.

TACHS-WE7-030 - Correct: D - Correct Answer: Without a corrected sign, a visitor could travel to a route that cannot be completed. - Explanation: The sentence explains the practical consequence of inaccurate temporary directions.

TACHS-WE7-031 - Correct: D - Correct Answer: After each drop, the disk was removed and inspected for cracks. - Explanation: The passive verb phrase requires the past participle "removed" after "was."

TACHS-WE7-032 - Correct: A - Correct Answer: Repeating each method several times would show whether the initial results are dependable rather than accidental. - Explanation: Choice A joins the proposed next step with its purpose in one complete sentence.

TACHS-WE7-033 - Correct: B - Correct Answer: while using the lowest mass of packing material - Explanation: The revision states the comparison naturally and identifies the material being measured.

TACHS-WE7-034 - Correct: D - Correct Answer: The equipment needs to be cleaned after each use. - Explanation: "Equipment" is a noncount singular noun and correctly takes "needs."

TACHS-WE7-035 - Correct: D - Correct Answer: The orchestra tuned while the audience entered. - Explanation: Choice D contains a complete independent clause with a properly attached dependent clause.

TACHS-WE7-036 - Correct: C - Correct Answer: We postponed the trip because it was raining. - Explanation: Choice C expresses the cause directly without redundancy.

TACHS-WE7-037 - Correct: B - Correct Answer: Line 4 - Explanation: Line 4 requires the subject pronoun "They," not the object pronoun "Them."

TACHS-WE7-038 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 requires "except," meaning other than, not the verb "accept."

MATHEMATICS ANSWERS

TACHS-M7-026 - Correct: B - Correct Answer: 35 - Explanation: $42 \div 6 \times 5 = 35$.

TACHS-M7-027 - Correct: A - Correct Answer: 12 - Explanation: Subtract $2x$ and 9 to get $x = 12$.

TACHS-M7-028 - Correct: B - Correct Answer: 17 - Explanation: The total is 85, and $85 \div 5 = 17$.

TACHS-M7-029 - Correct: C - Correct Answer: 80 cm² - Explanation: Area = $\frac{1}{2} \times 10 \times 16 = 80$ square centimeters.

TACHS-M7-030 - Correct: C - Correct Answer: 3.6 kg - Explanation: $450 \times 8 = 3,600$ grams, which equals 3.6 kilograms.

TACHS-M7-031 - Correct: D - Correct Answer: 59 - Explanation: 59 has no positive factors other than 1 and 59.

TACHS-M7-032 - Correct: C - Correct Answer: $8y - 7$ - Explanation: Distribute and combine constants: $8y - 12 + 5 = 8y - 7$.

TACHS-M7-033 - Correct: B - Correct Answer: $\frac{7}{12}$ - Explanation: Seven of the twelve marbles are red or green, so the probability is $\frac{7}{12}$.

TACHS-M7-034 - Correct: A - Correct Answer: 105° - Explanation: Quadrilateral angles total 360° ; $360 - 255 = 105^\circ$.

TACHS-M7-035 - Correct: D - Correct Answer: 144 - Explanation: The rate is 12 pages per minute, and $12 \times 12 = 144$.

TACHS-M7-036 - Correct: B - Correct Answer: $4\frac{1}{4}$ - Explanation: 0.25 equals $\frac{1}{4}$, so $4.25 = 4\frac{1}{4}$.

TACHS-M7-037 - Correct: C - Correct Answer: 19 - Explanation: $h(3) = 2(3^2) + 1 = 18 + 1 = 19$.

ABILITY ANSWERS

TACHS-A7-025 - Correct: D - Correct Answer: $\diamond \times \triangle$ - Explanation: Swapping the first and third symbols gives $\diamond \times \triangle$.

TACHS-A7-026 - Correct: C - Correct Answer: $\square \square \square \square$ - Explanation: Doubling two open squares produces four open squares.

TACHS-A7-027 - Correct: B - Correct Answer: Left $\triangle$; right $\diamond$ - Explanation: Both filled shapes change to their open forms without changing left-right positions.

TACHS-A7-028 - Correct: C - Correct Answer: $\square \backslash$ - Explanation: Rotating / clockwise gives $\backslash$, and the line moves to the right of $\square$.

TACHS-A7-029 - Correct: A - Correct Answer: $\{+, \star\}$ - Explanation: Removing the shared $\times$ leaves $+$ and $\star$.

TACHS-A7-030 - Correct: B - Correct Answer: Two holes on the left side, mirrored across the horizontal centerline - Explanation: The horizontal unfold creates a matching hole on the same left side at the reflected height.

TACHS-A7-031 - Correct: C - Correct Answer: Four holes in corresponding positions across both centerlines - Explanation: Two perpendicular folds reflect the punch into four corresponding locations.

TACHS-A7-032 - Correct: C - Correct Answer: One complete circular opening centered on the diagonal - Explanation: The semicircular cut on the fold reflects to form one complete circle centered on that diagonal.

TACHS-A7-033 - Correct: A - Correct Answer: Four holes forming two horizontal pairs - Explanation: Each off-fold punch reflects horizontally, creating two horizontal pairs and four holes.

TACHS-A7-034 - Correct: D - Correct Answer: □ inside △; × outside - Explanation: One open shape is inside a different open shape, with × outside.

TACHS-A7-035 - Correct: A - Correct Answer: ■ □ ▲ △ - Explanation: Each sequence contains filled-open pairs of two different shapes.

TACHS-A7-036 - Correct: C - Correct Answer: Top +; left □; right △ - Explanation: A plus is above two different open shapes.

TACHS-A7-037 - Correct: A - Correct Answer: ◇ becomes ◇ ◇ ◇ - Explanation: One open symbol becomes three identical copies of itself.

Exam Part D

Filename: tachs_full_exam_07_part_d.txt

Exam Family: TACHS

Exam Number: 7

Exam Part: D

Question Count: 50

QUESTIONS

SECTION 1 - READING

STIMULUS START

Stimulus ID: TACHS-R7-S10

Section: Reading

Stimulus Type: Literary Passage

Title: The Uneven Kite

Text:

Jun had built a diamond kite for the neighborhood field day. During its first flight, a sudden gust tore a narrow strip from the right edge. He patched the paper, but the kite kept turning right and diving.

His first solution was to add a longer tail. The extra weight slowed the kite without correcting the turn. Jun then held the kite by its bridle and compared the two sides. The patch was strong, but it made the right side slightly heavier.

Jun could not remove the patch without reopening the tear. Instead, he tied a short ribbon to the left side of the tail frame. He tested the kite, shortened the ribbon, and shifted its knot outward. On the third attempt, the kite climbed steadily. The ribbon did not match the original design, yet it balanced the pull created by the repair.

At field day, someone asked why the kite had a ribbon on only one side. Jun began to say it was covering a mistake, then stopped. The ribbon covered nothing. It showed the adjustment that allowed the kite to fly. He left it visible, a small record of the difference between restoring an object's appearance and restoring what it could do.

STIMULUS END

TACHS-R7-039

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S10

Prompt:

What makes the kite's right side slightly heavier?

- A) The ribbon on the left side
- B) The longer tail
- C) The bridle string
- D) The patch over the torn paper

TACHS-R7-040

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S10

Prompt:

Why does Jun move the ribbon's knot outward?

- A) To adjust how strongly the ribbon balances the kite
- B) To hide the patch from view
- C) To make the kite turn farther right
- D) To attach the ribbon to the bridle

TACHS-R7-041

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S10

Prompt:

Why does the author state that the ribbon "covered nothing"?

- A) To show that Jun forgot where the tear was
- B) To emphasize that the visible addition served a functional purpose rather than hiding damage
- C) To explain that the ribbon was too narrow to see
- D) To suggest that the ribbon should have been removed before field day

TACHS-R7-042

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S10

Prompt:

As used in the final paragraph, "record" most nearly means

- A) a measured flight speed.
- B) visible evidence of what happened.
- C) the highest score achieved.
- D) a musical recording.

STIMULUS START

Stimulus ID: TACHS-R7-S11

Section: Reading

Stimulus Type: Scientific Passage

Title: How Owl Wings Quiet the Air

Text:

Many owls approach prey with remarkably little flight noise. Their silence does not come from one feature alone. Several structures on their wings alter how air moves and how sound is produced.

Along the front edge of an owl's outer flight feathers are comb-like projections. These break a large stream of turbulent air into smaller currents. The upper feather surfaces also have a soft, velvety texture that helps reduce friction noise. At the trailing edge, flexible fringes allow air to leave the wing less abruptly.

These features involve trade-offs. A very soft, fringed feather may be less durable or less efficient for fast flight than a smooth, stiff feather. Owls generally benefit from quiet, controlled movement while hunting at moderate speeds. Birds that travel long distances rapidly may have different wing

structures because their needs differ.

Engineers study owl feathers when designing quieter fans, turbines, and aircraft parts. A machine does not need to resemble a bird. Designers can test the underlying principles, such as dividing turbulent airflow or softening a sharp edge. Biological inspiration is most useful when the function of a structure is understood, not merely its appearance copied.

STIMULUS END

TACHS-R7-043

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S11

Prompt:

Where are the comb-like projections located?

- A) Beneath the trailing edge of the tail
- B) On the owl's beak
- C) Along the front edge of outer flight feathers
- D) Inside the wing bones

TACHS-R7-044

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S11

Prompt:

Which statement best summarizes the passage?

- A) Several feather features reduce flight noise and offer principles that engineers can adapt.
- B) Owls are silent because they never create turbulent air.
- C) All birds would fly better with soft, fringed feathers.
- D) Machines must copy the exact shape of an owl to become quieter.

TACHS-R7-045

Section: Reading

Type: mcq

Category: Reading

Skill: Vocabulary in Context

Stimulus ID: TACHS-R7-S11

Prompt:

As used in the third paragraph, "trade-offs" most nearly means

- A) measurements taken during flight.
- B) features found in every bird.
- C) situations in which gaining one benefit may involve giving up another.
- D) exchanges of feathers between animals.

TACHS-R7-046

Section: Reading

Type: mcq

Category: Reading

Skill: Evaluative Comprehension, Author's Purpose, or Tone

Stimulus ID: TACHS-R7-S11

Prompt:

Why does the author compare owls with birds that travel rapidly?

- A) To prove that rapid birds make no sound
- B) To show that every bird has identical flight needs
- C) To argue that owls cannot control their movement
- D) To explain why wing structures reflect different functional priorities.

STIMULUS START

Stimulus ID: TACHS-R7-S12

Section: Reading

Stimulus Type: Historical-Style Passage

Title: Books Traveling by Post

Text:

Before digital catalogs and electronic books, some libraries served distant readers by mail. A person living far from a branch could request a title from a printed catalog. Library workers wrapped the book, recorded the loan, and sent it through the postal system.

Packaging mattered. A container had to protect the book while remaining light enough to mail affordably. Some libraries used reusable cases with labels that could be reversed for the return trip. Instructions reminded borrowers to keep the packaging and send the book back by a specified date.

Mail service expanded access, but it also changed selection. A reader could not browse shelves or examine several books before choosing. Clear catalog descriptions and correspondence with librarians became important. Delays caused by distance or weather also meant that loans often lasted longer than in-person loans.

These programs required coordination among catalogs, postal routes, packaging, records, and readers. The traveling book was only the visible part of that system. By treating distance as a practical problem to be managed, mail libraries extended a familiar public service beyond the walls of a building.

STIMULUS END

TACHS-R7-047

Section: Reading

Type: mcq

Category: Reading

Skill: Factual Comprehension, Detail, or Textual Evidence

Stimulus ID: TACHS-R7-S12

Prompt:

Why were some mailing labels reversible?

- A) Postal workers could remove the due date.
- B) Borrowers could change the book's title.
- C) The same case could be used for the return trip.
- D) Libraries could avoid keeping loan records.

TACHS-R7-048

Section: Reading

Type: mcq

Category: Reading

Skill: Interpretative Comprehension or Inference

Stimulus ID: TACHS-R7-S12

Prompt:

Why were clear catalog descriptions especially important for mail borrowers?

- A) They were required to write the catalog themselves.
- B) They received every available title automatically.
- C) They could not browse the shelves before selecting a book.
- D) They could return books without packaging.

TACHS-R7-049

Section: Reading

Type: mcq

Category: Reading

Skill: Main Idea, Central Thought, or Summary

Stimulus ID: TACHS-R7-S12

Prompt:

What is the central idea of the passage?

- A) Postal service made library records unnecessary.
- B) Mail libraries coordinated several systems to extend book access across distance.
- C) Readers preferred packaging to the books inside it.
- D) Borrowing by mail was faster than visiting a local branch.

TACHS-R7-050

Section: Reading

Type: mcq

Category: Reading

Skill: Organization, Text Structure, or Literary Technique

Stimulus ID: TACHS-R7-S12

Prompt:

How is the passage mainly organized?

- A) A fictional reader describes one favorite novel.
- B) A service is introduced, followed by its practical requirements, limitations, and broader significance.
- C) Mail routes are listed by geographic direction.
- D) Two postal companies argue about prices.

SECTION 2 - WRITTEN EXPRESSION

STIMULUS START

Stimulus ID: TACHS-WE7-S07

Section: Written Expression

Stimulus Type: Revision Passage

Title: Choosing Shade for a Bus Stop

Text:

[1] Residents have requested more shade at the bus stop beside Oak Avenue. [2] Before recommending a structure, our planning group observed the site at four times of day. [3] We recorded where shadows fell, how many passengers waited, and whether parked vehicles blocked the curb. [4] A large tree stands nearby, its shadow reaches the shelter only in early morning. [5] At midday, the waiting area receives direct sunlight for nearly three hours. [6] The group recommends a narrow overhead canopy positioned so it does not interfere with buses. [7] We should also measure

the site again in winter, when the Sun follows a lower path. [8] Seasonal observations will help confirm that the canopy works throughout the year.

STIMULUS END

TACHS-WE7-039

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S07

Prompt:

Which sentence best explains why the group observed the site at several times?

- A) Sentence 1
- B) Sentence 8
- C) Sentence 5
- D) Sentence 2

TACHS-WE7-040

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S07

Prompt:

Which revision corrects the error in sentence 4?

- A) A large tree stands nearby its shadow reaches the shelter, only in early morning.
- B) A large tree stands nearby, but its shadow reaches the shelter only in early morning.
- C) A large tree standing nearby, its shadow only in early morning.
- D) Although a large tree stands nearby and its shadow reaches the shelter.

TACHS-WE7-041

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S07

Prompt:

Which revision of sentence 6 is most precise?

- A) The group recommends putting a shade thing somewhere over there.
- B) A canopy should be kind of narrow and should not cause problems.
- C) It is recommended that something overhead be positioned appropriately.
- D) The group recommends a 1.5-meter-deep canopy above the marked waiting zone, set back from the curb.

TACHS-WE7-042

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Organization and Development

Stimulus ID: TACHS-WE7-S07

Prompt:

Which sentence would best follow sentence 5?

- A) The bus route began operating twelve years ago.
- B) Oak Avenue contains several grocery stores.
- C) This period overlaps with the busiest afternoon boarding times.
- D) Some passengers carry umbrellas in the rain.

STIMULUS START

Stimulus ID: TACHS-WE7-S08

Section: Written Expression

Stimulus Type: Revision Passage

Title: Recording Community Memories

Text:

[1] Our local history group plans to record short interviews with longtime residents. [2] Before each interview, participants will receive a written explanation of the project and how recordings may be used. [3] Interviewers will ask permission before beginning and again before sharing any excerpt publicly. [4] Questions should invite detailed memories without suggesting a preferred answer. [5] For example, "What do you remember about shopping on this street?" is more open than "Was the old market better?" [6] Each recording and consent form will be given matching identification numbers. [7] Personal contact information are stored separately from the interview files. [8] These procedures respect participants while creating an organized collection for future research.

STIMULUS END

TACHS-WE7-043

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: TACHS-WE7-S08

Prompt:

Which revision corrects the error in sentence 7?

- A) Personal contact information is stored separately from the interview files.
- B) Personal contact informations are stored separately from the interview files.
- C) Personal contact information were stored separately from the interview files.
- D) Personal contact information be stored separately from the interview files.

TACHS-WE7-044

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Sentence Structure

Stimulus ID: TACHS-WE7-S08

Prompt:

Which revision most effectively combines sentences 2 and 3?

- A) Participants with a written explanation, and permission before, recordings shared publicly.
- B) The project explains uses because participants and permission again publicly.
- C) Before each interview and sharing excerpts, a written explanation being received.

D) After receiving a written explanation of the project and its possible uses, participants will be asked for permission before recording and before public sharing.

TACHS-WE7-045

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Clarity, Effectiveness, or Appropriateness

Stimulus ID: TACHS-WE7-S08

Prompt:

Which phrase best replaces “suggesting a preferred answer” in sentence 4?

- A) getting people to say stuff the interviewer wants
- B) making the answer preferred by a suggestion
- C) leading the participant toward a particular response
- D) putting an answer into a question somehow

TACHS-WE7-046

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) The interview differed from the earlier one.
- B) The interview differed with the earlier one.
- C) The interview was different than the earlier one.
- D) The interview was differing to the earlier one.

TACHS-WE7-047

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Grammar and Usage

Stimulus ID: None

Prompt:

Which sentence is written correctly?

- A) Whom left the package at the desk?
- B) To who should I address the letter?
- C) Who called the manager yesterday?
- D) Whom should receive the award?

TACHS-WE7-048

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The mechanic inspected both wheels.

Line 2: One tire had lost some air overnight.

Line 3: He filled it, and then checked the pressure.

Line 4: The bicycle was ready by noon.

A) Line 3

B) Line 2

C) Line 1

D) Line 4

TACHS-WE7-049

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: We set the seedlings beside the window.

Line 2: Each pot received an equal amount of water.

Line 3: New leaves appeared within a week.

Line 4: The tallest plant grew more quicker than the others.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

TACHS-WE7-050

Section: Written Expression

Type: mcq

Category: Written Expression

Skill: Error Identification

Stimulus ID: None

Prompt:

Which numbered line contains an error in standard written US English?

Line 1: The museum opens at nine o'clock.

Line 2: Visitors may purchase tickets at the entrance.

Line 3: Large bags must be left in a locker.

Line 4: Photography is permitted, accept in the special exhibit.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

SECTION 3 - MATHEMATICS

TACHS-M7-038

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

What is 1.5×0.24 ?

- A) 0.036
- B) 36
- C) 3.6
- D) 0.36

TACHS-M7-039

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

The values are 6, 8, 8, 10, 13, and 15. What is the median?

- A) 8
- B) 9
- C) 10
- D) 11

TACHS-M7-040

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A circle has circumference 31.4 centimeters. Using $\pi = 3.14$, what is its diameter?

- A) 10 cm
- B) 8 cm
- C) 5 cm
- D) 15.7 cm

TACHS-M7-041

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

Solve $4 - 3x = 19$.

- A) -3
- B) -5
- C) 3
- D) 5

TACHS-M7-042

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A tank holds 6.4 liters. How many 800-milliliter containers can be filled?

- A) 6
- B) 7
- C) 9
- D) 8

TACHS-M7-043

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which ratio is equivalent to 18:30?

- A) 3:5
- B) 2:5
- C) 5:6
- D) 9:10

TACHS-M7-044

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A spinner has five equal sections labeled A, A, B, C, and D. What is the probability of not landing on A?

- A) $\frac{1}{5}$
- B) $\frac{3}{5}$
- C) $\frac{2}{5}$
- D) $\frac{4}{5}$

TACHS-M7-045

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A rectangular garden is 18 m by 12 m. A path 1 m wide runs inside all four edges. What is the area of the remaining inner rectangle?

A) 180 m²

B) 160 m²

C) 192 m²

D) 216 m²

TACHS-M7-046

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Algebraic Patterns and Connections

Stimulus ID: None

Prompt:

The first term of a sequence is 4. Each new term is 3 more than twice the previous term. What is the third term?

A) 14

B) 19

C) 22

D) 25

TACHS-M7-047

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Measurement

Stimulus ID: None

Prompt:

A rectangular floor is 4.5 m long and 3 m wide. What is its area?

A) 7.5 m²

B) 27 m²

C) 15 m²

D) 13.5 m²

TACHS-M7-048

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Number Sense and Operations

Stimulus ID: None

Prompt:

Which value is greatest?

A) $\frac{7}{9}$

B) 0.79

- C) 0.8
D) 78%

TACHS-M7-049

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Data Analysis, Probability, or Statistics

Stimulus ID: None

Prompt:

A student's scores are 70, 76, 82, and 88. What score is needed for a five-score mean of 80?

- A) 80
B) 82
C) 84
D) 86

TACHS-M7-050

Section: Mathematics

Type: mcq

Category: Mathematics

Skill: Geometry

Stimulus ID: None

Prompt:

A right rectangular prism has volume 180 cm^3 , length 10 cm, and width 6 cm. What is its height?

- A) 3 cm
B) 4 cm
C) 6 cm
D) 2 cm

SECTION 4 - ABILITY

TACHS-A7-038

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Matrices

Stimulus ID: None

Prompt:

Complete the matrix. In each row, the right cell moves the center symbol to both ends.

Row 1: $\bigcirc + \square \mid + \bigcirc \square +$

Row 2: $\triangle \times \diamond \mid ?$

- A) $\times \triangle \diamond \times$
B) $\triangle \times \times \diamond$
C) $\times \triangle \times \diamond$
D) $\triangle \diamond \times \times$

TACHS-A7-039

Section: Ability

Type: mcq
Category: Ability
Skill: Figure Matrices
Stimulus ID: None
Prompt:

Complete the matrix. Down each column, a filled shape becomes open and moves from top to bottom.

Top-left: Top ●

Bottom-left: Bottom ○

Top-right: Top ■

Bottom-right: ?

A) Top □

B) Bottom □

C) Bottom ■

D) Top ■

TACHS-A7-040

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Matrices
Stimulus ID: None
Prompt:

Complete the matrix. Across each row, the second cell contains one fewer symbol and changes open to filled.

Row 1: ○ ○ ○ | ● ●

Row 2: □ □ □ □ | ?

A) □ □ □

B) ■ ■

C) ■ ■ ■

D) ■ ■ ■ ■

TACHS-A7-041

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Matrices
Stimulus ID: None
Prompt:

Complete the matrix. Across each row, the arrow turns 45° counterclockwise and the shape changes from filled to open.

Row 1: → ● | / ○

Row 2: ↓ ■ | ?

A) / □

B) \ □

C) \ ■

D) → ☐

TACHS-A7-042

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded from right to left. Three holes are punched in a vertical column on the folded paper, away from all edges. What appears after unfolding?

- A) Three holes on the vertical fold
- B) Six holes forming three horizontal mirrored pairs
- C) Six holes in one vertical line
- D) Twelve holes around the edges

TACHS-A7-043

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A rectangle is folded from top to bottom. A notch is cut from the folded paper's right outer edge, away from corners. What appears after unfolding?

- A) One notch at the center
- B) Two notches on opposite side edges
- C) Two matching notches on the right edge, mirrored vertically
- D) Four notches at the corners

TACHS-A7-044

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded left to right and then top to bottom. A triangular cut is made on the folded paper's vertical outer edge, away from corners. What appears after unfolding?

- A) Four cuts, two on each opposite vertical edge
- B) Four cuts at the center
- C) Two cuts on the vertical centerline
- D) Eight cuts on all four edges

TACHS-A7-045

Section: Ability

Type: mcq

Category: Ability

Skill: Paper Folding

Stimulus ID: None

Prompt:

A square is folded along the diagonal from upper-left to lower-right. One hole is punched near each of two different outer edges of the folded triangle, away from corners. What appears after unfolding?

- A) Two holes total
- B) Four holes on one edge
- C) Four holes forming two pairs mirrored across the diagonal
- D) Eight holes around the center

TACHS-A7-046

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following arrangements share one rule:

- 1. ○ above ●
- 2. □ above ■
- 3. △ above ▲

Which choice follows the same rule?

- A) ○ above ■
- B) ◆ above ◇
- C) ☆ above ○
- D) ◇ above ◆

TACHS-A7-047

Section: Ability

Type: mcq

Category: Ability

Skill: Figure Classification

Stimulus ID: None

Prompt:

The following groups share one rule:

- 1. + + ○ ×
- 2. + + □ ×
- 3. + + △ ×

Which choice follows the same rule?

- A) + ◇ + ×
- B) + + ◆ ×
- C) + + ◇ ×
- D) × + + ◇

TACHS-A7-048

Section: Ability

Type: mcq

Category: Ability
Skill: Figure Classification
Stimulus ID: None
Prompt:

The following figures share one rule:

1. Outer □; ○ at bottom-right inside
2. Outer ◇; △ at bottom-right inside
3. Outer ○; ☆ at bottom-right inside

Which choice follows the same rule?

- A) Outer ■; ○ at bottom-right
B) Outer △; □ at bottom-right
C) Outer □; ■ at bottom-right
D) Outer □; ○ at top-right

TACHS-A7-049

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Classification
Stimulus ID: None
Prompt:

The following sequences share one rule:

1. ↑ → ↓ ←
2. → ↓ ← ↑
3. ↓ ← ↑ →

Which choice follows the same rule?

- A) ← ↓ → ↑
B) ← ↑ → ↓
C) → ↑ ← ↓
D) ↑ ← ↓ →

TACHS-A7-050

Section: Ability
Type: mcq
Category: Ability
Skill: Figure Classification
Stimulus ID: None
Prompt:

The following groups share one rule:

1. ○ □ ○ □ ○
2. △ ◇ △ ◇ △
3. ☆ ○ ☆ ○ ☆

Which choice follows the same rule?

- A) □ □ △ △ □

B) ■ △ ■ △ ■

C) □ △ △ □ △

D) □ △ □ △ □

ANSWER KEY + EXPLANATIONS

READING ANSWERS

TACHS-R7-039 - Correct: D - Correct Answer: The patch over the torn paper - Explanation: Jun observes that the strong patch adds weight to the right side.

TACHS-R7-040 - Correct: A - Correct Answer: To adjust how strongly the ribbon balances the kite - Explanation: Changing the ribbon's position lets Jun fine-tune its balancing effect during testing.

TACHS-R7-041 - Correct: B - Correct Answer: To emphasize that the visible addition served a functional purpose rather than hiding damage - Explanation: The ribbon openly records the adjustment that restored flight instead of concealing the repair.

TACHS-R7-042 - Correct: B - Correct Answer: visible evidence of what happened. - Explanation: The ribbon remains as visible evidence of the change that made the kite functional.

TACHS-R7-043 - Correct: C - Correct Answer: Along the front edge of outer flight feathers - Explanation: The second paragraph places the comb-like projections on the front edge of the outer flight feathers.

TACHS-R7-044 - Correct: A - Correct Answer: Several feather features reduce flight noise and offer principles that engineers can adapt. - Explanation: The passage explains multiple noise-reducing features, their trade-offs, and their possible engineering use.

TACHS-R7-045 - Correct: C - Correct Answer: situations in which gaining one benefit may involve giving up another. - Explanation: Quiet feather structures may reduce durability or speed efficiency, illustrating competing benefits.

TACHS-R7-046 - Correct: D - Correct Answer: To explain why wing structures reflect different functional priorities. - Explanation: The comparison shows that quiet hunting and fast long-distance travel favor different feather characteristics.

TACHS-R7-047 - Correct: C - Correct Answer: The same case could be used for the return trip. - Explanation: A reversible label allowed the reusable package to be addressed back to the library.

TACHS-R7-048 - Correct: C - Correct Answer: They could not browse the shelves before selecting a book. - Explanation: Because distant readers could not inspect books in person, descriptions guided their choices.

TACHS-R7-049 - Correct: B - Correct Answer: Mail libraries coordinated several systems to extend book access across distance. - Explanation: The passage emphasizes the coordinated logistics that allowed libraries to serve distant readers.

TACHS-R7-050 - Correct: B - Correct Answer: A service is introduced, followed by its practical requirements, limitations, and broader significance. - Explanation: The passage explains how mail lending worked, what it required, its constraints, and its purpose.

WRITTEN EXPRESSION ANSWERS

TACHS-WE7-039 - Correct: D - Correct Answer: Sentence 2 - Explanation: Sentence 2 introduces repeated observation before a recommendation, and later details show how conditions vary.

TACHS-WE7-040 - Correct: B - Correct Answer: A large tree stands nearby, but its shadow reaches the shelter only in early morning. - Explanation: A comma and coordinating conjunction correctly join the two contrasting independent clauses.

TACHS-WE7-041 - Correct: D - Correct Answer: The group recommends a 1.5-meter-deep canopy above the marked waiting zone, set back from the curb. - Explanation: Choice D gives specific dimensions and placement while preserving bus clearance.

TACHS-WE7-042 - Correct: C - Correct Answer: This period overlaps with the busiest afternoon boarding times. - Explanation: The sentence connects the midday sunlight problem with the number of people affected.

TACHS-WE7-043 - Correct: A - Correct Answer: Personal contact information is stored separately from the interview files. - Explanation: "Information" is a singular noncount noun and takes the verb "is."

TACHS-WE7-044 - Correct: D - Correct Answer: After receiving a written explanation of the project and its possible uses, participants will be asked for permission before recording and before public sharing. - Explanation: Choice D clearly combines the explanation and two stages of permission without a misplaced modifier.

TACHS-WE7-045 - Correct: C - Correct Answer: leading the participant toward a particular response - Explanation: The revision precisely explains the problem with leading interview questions.

TACHS-WE7-046 - Correct: A - Correct Answer: The interview differed from the earlier one. - Explanation: The standard expression is "differed from."

TACHS-WE7-047 - Correct: C - Correct Answer: Who called the manager yesterday? - Explanation: "Who" is correct because it serves as the subject of the verb "called."

TACHS-WE7-048 - Correct: A - Correct Answer: Line 3 - Explanation: Line 3 contains an unnecessary comma between two verbs sharing the subject "He."

TACHS-WE7-049 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 uses a double comparative; "more" should be removed.

TACHS-WE7-050 - Correct: D - Correct Answer: Line 4 - Explanation: Line 4 requires "except," meaning excluding, rather than "accept."

MATHEMATICS ANSWERS

TACHS-M7-038 - Correct: D - Correct Answer: 0.36 - Explanation: $15 \times 24 = 360$, and three decimal places give $0.360 = 0.36$.

TACHS-M7-039 - Correct: B - Correct Answer: 9 - Explanation: The middle values are 8 and 10, so the median is 9.

TACHS-M7-040 - Correct: A - Correct Answer: 10 cm - Explanation: Diameter = circumference $\div \pi = 31.4 \div 3.14 = 10$ centimeters.

TACHS-M7-041 - Correct: B - Correct Answer: -5 - Explanation: Subtract 4 to get $-3x = 15$, so $x = -5$.

TACHS-M7-042 - Correct: D - Correct Answer: 8 - Explanation: 6.4 liters = 6,400 milliliters, and $6,400 \div 800 = 8$.

TACHS-M7-043 - Correct: A - Correct Answer: 3:5 - Explanation: Dividing both terms by 6 gives 3:5.

TACHS-M7-044 - Correct: B - Correct Answer: $3/5$ - Explanation: Three of the five sections are not A, so the probability is $3/5$.

TACHS-M7-045 - Correct: B - Correct Answer: 160 m^2 - Explanation: The inner dimensions are 16 m by 10 m, so its area is 160 square meters.

TACHS-M7-046 - Correct: D - Correct Answer: 25 - Explanation: The second term is 11, and the third is $2(11) + 3 = 25$.

TACHS-M7-047 - Correct: D - Correct Answer: 13.5 m^2 - Explanation: Area = $4.5 \times 3 = 13.5$ square meters.

TACHS-M7-048 - Correct: C - Correct Answer: 0.8 - Explanation: $7/9 \approx 0.778$, $78\% = 0.78$, and 0.8 is greatest.

TACHS-M7-049 - Correct: C - Correct Answer: 84 - Explanation: A total of 400 is needed; current total is 316, so the fifth score must be 84.

TACHS-M7-050 - Correct: A - Correct Answer: 3 cm - Explanation: Height = $180 \div (10 \times 6) = 3$ centimeters.

ABILITY ANSWERS

TACHS-A7-038 - Correct: A - Correct Answer: $\times \triangle \diamond \times$ - Explanation: The center $\times$ is copied to both ends of the unchanged $\triangle \diamond$ pair.

TACHS-A7-039 - Correct: B - Correct Answer: Bottom $\square$ - Explanation: The filled square becomes open and moves from top to bottom.

TACHS-A7-040 - Correct: C - Correct Answer: $\blacksquare \blacksquare \blacksquare$ - Explanation: Four open squares become three filled squares.

TACHS-A7-041 - Correct: B - Correct Answer: $\setminus \square$ - Explanation: Turning $\downarrow$ 45° counterclockwise gives $\setminus$, and $\blacksquare$ becomes open $\square$.

TACHS-A7-042 - Correct: B - Correct Answer: Six holes forming three horizontal mirrored pairs - Explanation: Each of the three punches reflects horizontally, producing three mirrored pairs and six holes.

TACHS-A7-043 - Correct: C - Correct Answer: Two matching notches on the right edge, mirrored vertically - Explanation: The horizontal unfold reflects the notch to another position on the same right edge.

TACHS-A7-044 - Correct: A - Correct Answer: Four cuts, two on each opposite vertical edge - Explanation: One unfold doubles the cut vertically and the other mirrors both to the opposite vertical edge.

TACHS-A7-045 - Correct: C - Correct Answer: Four holes forming two pairs mirrored across the diagonal - Explanation: Each of the two off-fold punches reflects once, producing two mirrored pairs.

TACHS-A7-046 - Correct: D - Correct Answer: ◇ above ◆ - Explanation: An open shape appears above its matching filled form.

TACHS-A7-047 - Correct: C - Correct Answer: + + ◇ × - Explanation: Two plus signs are followed by one open shape and then ×.

TACHS-A7-048 - Correct: B - Correct Answer: Outer △; □ at bottom-right - Explanation: An open outer shape contains a different open shape at the bottom-right.

TACHS-A7-049 - Correct: B - Correct Answer: ← ↑ → ↓ - Explanation: Each sequence turns 90° clockwise at every step; ←, ↑, →, ↓ follows that rule.

TACHS-A7-050 - Correct: D - Correct Answer: □ △ □ △ □ - Explanation: Each group alternates two open shapes and begins and ends with the first shape.